The Defensible AI Framework Registry

Canonical Names, Definitions and Relationships for the Governed Production AI Discipline

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Trademark notice. MESA Framework™, MESA MRM Framework™, AI Data Governance Framework™, AI Vendor Risk Framework (AVRF)™, AI Incident Response Protocol (AIRP)™, Five-Gate Deployment Model™, Sharia AI Compliance Framework (SACF)™, Cross-Border AI Architecture Patterns™ and CADRE™ are trademarks of the author and iSystematic Inc. **CADRE is new at this version**, and no trademark clearance search has been completed for it; 4.7.4 records that as open. The Boundary Invariant, the BOE Declaration, PEVG and PARA remain deliberately unmarked, for the reasons recorded at 2.3. Where text is licensed CC BY 4.0, the licence grants no rights in the marks.

Abstract

A body of frameworks that shares an author does not thereby share an architecture. Nine frameworks for governing artificial intelligence in regulated institutions were developed and published separately, each answering a question a supervisor or a board had asked. Read together they carried three defects that no individual framework could show: three separate names described one subject, two instruments measured the same maturity, and nothing stated why any of them belonged in the same system as the others.

This registry is the instrument that fixes those defects and the authoritative record of the result. It states one architecture for the nine, assigns each framework a stable identifier and a canonical name, and publishes a normative relationship register naming which framework supplies the artifact, evidence or authority another depends on.

The architecture rests on two propositions that are already published and separately citable. The **MESA Framework™**, a four-altitude diagnostic model for institutional AI governance, supplies the frame: governance is an alignment condition across those altitudes, so maturity is reported as a per-altitude profile rather than as a single grade. The **Boundary Invariant** supplies the engineering law: a boundary is a clause the optimizer may not cross, and everything else is optimization. Between them they fix the two directions in which the composed system is read. Authority flows down, from the altitude that binds to the control that executes. Evidence flows up, from the record a control writes to the profile the institution reports.

Each of the nine entries carries a definition, a purpose, an altitude or boundary class, inputs, outputs, one accountable role, an evidence requirement, its relationships, its limitations, the condition that would falsify it, and its change history. The entry schema is normative and is published in machine-readable form as `registry-v2.0.json`, validated against `registry-schema-v2.0.json`, so that a machine interface and this document cannot drift apart without one of them failing a test.

The registry also serves as the dated instrument of a consolidation. Three marks are recorded as consolidated into one framework, one instrument is recorded as superseded, one framework is recorded as renamed, and one is recorded as repositioned from peer framework to worked instance of the Boundary Invariant. Twelve asserted marks became eight marks and one deliberately unmarked framework at version 1.0, and nine marks at this version, where the unmarked framework is renamed and marked. **All nine entries now carry a deposited specification**, which closes the state version 1.0 recorded as its expected change.

The contribution is architectural rather than empirical. The registry describes a composition. No institution unconnected to the author has been observed operating the composed set, no framework in it carries an independent evaluation, and Section 10 states what would falsify the composition as distinct from what would falsify any framework within it. It is a specification, not a certification scheme, and no conformity assessment body operates against it.

Keywords: AI governance; framework registry; governance architecture; model risk management; vendor risk; incident response; deployment gates; data governance; Sharia governance; cross-border AI; evidence architecture; regulated industries.

Notation

Normative terms

This specification uses **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** as the normative verbal forms.

- **MUST** indicates a requirement. An entry, implementation or citation that does not meet it is not conformant with this registry.
- **SHOULD** indicates a recommendation. Departure requires a recorded rationale.
- **MAY** indicates a permission.

The two specifications this registry references most closely, the MESA Framework specification [1] and A Pattern Language for Production LLM Platforms [2], use the verbal forms of ISO/IEC Directives Part 2, in which **shall** carries the meaning assigned to MUST here. The correspondence is exact. No requirement changes meaning when a reader moves between the documents.

Defined terms

Framework. A named, reusable structure that decomposes a governance or engineering problem into parts, assigns accountability for each part, and specifies what evidence each part produces. In this registry a framework is the unit of entry.

Registry entry. The record this specification makes of one framework: the fifteen fields listed at 3.1, of which the first is its identifier and the second its canonical name. The entry is the authoritative statement of what the framework is called and how it relates to the others. It is not the framework's specification and does not replace one.

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Canonical name. The single name under which a framework is cited. Where a framework has been published under a different name, the former name is recorded at Appendix A and remains valid for citation of the earlier work.

Altitude. One of the four governance levels defined by the MESA Framework: the Regulatory Floor, the Strategic Compass, Operational Machinery and the Technical Substrate.

Boundary class. One of the five classes defined by A Pattern Language for Production LLM Platforms, covering data, authority, decision, resource and process constraints.

Authority source. A body whose determinations bind the institution and are satisfied rather than traded against likelihood and impact. A financial supervisor is an authority source. A Sharia Supervisory Board, where it holds binding approval power, is an authority source.

Evidence artifact. A record written so that a decision can be reconstructed by a party who was not present when it was taken, as distinguished from a log written for debugging.

Institution. The accountable legal entity operating the AI systems the frameworks govern. **Regulated institution** means one whose ability to operate is conditioned on its ability to reconstruct and defend a decision to an external authority.

Spelling of Sharia

The canonical name of the eighth framework is the **Sharia AI Compliance Framework (SACF)™**, spelled as first asserted, and that spelling MUST NOT be altered in citation. **Sharia** is used throughout, and the body of authority is written **Sharia Supervisory Board**. Where a cited instrument spells it otherwise, the title is quoted as its publisher printed it, so *Sharī'ah* and *Shari'ah* appear inside references to IFSB and AAOIFI instruments and nowhere else. The difference is orthographic only and denotes one subject.

Identifiers

Every framework in this registry carries a stable identifier of the form **REG-nn**. **The identifiers are normative and MUST NOT be renumbered.** A framework retired from the registry retains its identifier, which is marked retired rather than reused. Canonical names are aliases for the identifier, so that a rename changes the name while the identifier holds the citation chain together.

Normative status of sections

Sections 2 through 8 are normative. Sections 1, 9, 10 and 11 are informative, except where they state a requirement using **MUST**.

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1. Introduction

1.1 The problem this registry solves

An institution adopting a governance framework asks three questions of it before it asks anything else. What is this called, so that a policy can name it. What does it require, so that a control can satisfy it. What else does it depend on, so that a programme can sequence it.

A single framework published on its own answers the first two and cannot answer the third. A body of frameworks published separately answers the third only by inference, and inference is what produces the failure this registry exists to prevent: two documents from the same author asserting the same relationship in incompatible words, each defensible in isolation, and no instrument that says which one governs.

That failure had already begun. Twelve trademarks were asserted on the copyright page of a single book [13]. Three of them described one subject, which is how the structure of a governance function, the assignment of its decision rights, and the rhythm on which it meets came to be three frameworks rather than three sections of one. Two of them scored maturity, one formally and one informally, so that every conversation about maturity had first to establish which instrument was in use. One of them was named for a technology stack while its content was a control taxonomy. And nothing anywhere stated the reason the twelve belonged to one system, which meant a reader could accept every framework individually and still be unable to compose them.

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1.2 What a registry is for

A registry is not a summary of the things it lists. A summary makes a long thing short. A registry makes a set of things addressable: it fixes their names so that they can be cited, fixes their relationships so that they can be sequenced, and fixes its own version so that a claim made against it can be dated.

The test of a registry is therefore not whether it describes its members accurately. It is whether a reader who has read none of the member documents can determine, from the registry alone, which framework applies to a given question, which other frameworks that one requires, what evidence it produces, who is accountable for it, and what would show it to be wrong.

1.3 Contribution

This specification contributes four things.

1. **One architecture for nine frameworks**, stated in Section 2, resting on two propositions that are separately published and independently citable rather than on this document's own assertion.
2. **Nine normative registry entries**, in Section 4, each carrying the same fifteen fields, so that the frameworks can be compared field against field rather than prose against prose.
3. **A normative relationship register**, in Section 5, naming every dependency between the nine as a typed edge, with a closure rule requiring that each dependency be asserted exactly once, in a direction fixed by its type, and its converse derived rather than authored.
4. **A machine-readable layer**, in Section 7, published as one JSON document with its schema, so that a machine interface can serve the registry without a second and divergent statement of the same facts being authored to do it.

The consolidation recorded in Section 6 is not a contribution of this specification. It is a decision taken elsewhere, of which this specification is the dated instrument.

1.4 Scope

This registry covers the nine frameworks of the **governance family**, listed at 4.10. It is calibrated to regulated institutions operating AI systems under at least one binding authority source.

It does not cover the **engineering family**, whose canonical source is A Pattern Language for Production LLM Platforms [2]. The relationship between the two families, and the precedence rule that resolves a disagreement between them, is stated at 2.6.

1.5 Non-goals

This specification is none of the following, and a reader who needs one of them needs a different document.

- **Not a specification of any framework it lists.** An entry states what a framework is, what it requires and how it relates. It does not state how to operate it. Where a full specification exists it is cited in the entry; where none exists the entry says so.
- **Not a certification scheme.** No conformity assessment body operates against this registry, and no institution can be certified against it. Section 8 defines the only two conformance claims the registry supports, and both are self-declared.
- **Not a statement of regulatory compliance.** Applying every framework in this registry does not establish compliance with any statute, supervisory expectation or standard. The frameworks are structures for producing and organizing evidence. Whether that evidence satisfies an authority is the authority's determination.
- **Not a maturity assessment.** MESA supplies the maturity instrument. This registry records that it does.
- **Not an empirical claim.** Section 10 states what has not been tested.

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2. The Architecture (Normative)

2.1 Two propositions, not one

A registry that asserted its own architecture would be the weakest possible instrument, because the only evidence for the architecture would be the document claiming it. This registry therefore composes two propositions that were published separately, before it, and that can be examined without reference to it.

Proposition one, from the MESA Framework [1]. Institutional AI governance is an alignment condition across four distinct altitudes. Maturity MUST therefore be reported as a per-altitude profile rather than as a single grade, because a single grade allows a strong regulatory posture to conceal a technical substrate that cannot hold it up.

Proposition two, the Boundary Invariant, from A Pattern Language for Production LLM Platforms [2]. A boundary is a clause the optimizer may not cross, and everything else is optimization. Where the two are written into one clause, improving either silently erodes the other.

The first proposition decomposes the institution. The second decomposes the control. Neither entails the other, and the registry needs both: without the first there is no account of who binds whom, and without the second there is no account of what a control is obliged to hold fixed.

2.2 MESA as the diagnostic frame

Every framework in this registry except REG-01 and REG-09 **instantiates exactly one MESA altitude**. The altitude is a normative field of the entry, not a description of it, and it determines three things: which accountable role the framework reports through, which evidence the framework contributes to the institution's maturity profile, and which other frameworks may bind it.

- The **Regulatory Floor** carries what binds the institution from outside. Its inhabitant in this registry is REG-08, which is the mechanism by which a second binding authority source enters the same machinery as the first.
- The **Strategic Compass** carries what the institution chooses. Its inhabitant is REG-07, which converts a board mandate into a standing structure, a set of decision rights and a cadence.
- **Operational Machinery** carries what the governance function performs. Its inhabitants are REG-02, REG-04, REG-05 and REG-06.
- The **Technical Substrate** carries what the system enforces at runtime. Its inhabitant is REG-03.

Two entries stand outside that rule, for opposite reasons. **REG-01 defines the altitudes** and therefore occupies none of them. **REG-09 sits outside them altogether**, under the Boundary Invariant rather than under an altitude, for the reason recorded at 6.5. In both cases the registry records the absence as a property of the entry rather than treating it as a missing field.

2.3 The Boundary Invariant as the engineering law

Where MESA says which altitude binds a control, the Boundary Invariant says what the control owes. Every control operated under any framework in this registry **MUST carry a BOE Declaration**: a statement of the **B**oundary it fixes, the **O**ptimizer it frees, and the **E**vidence proving the boundary held.

The BOE Declaration is what turns the nine into one system rather than nine systems with one author. Two frameworks that produce evidence in the same shape can be joined; two that do not are merely adjacent. A gate record under REG-06, a validation record under REG-02, a residency attestation under REG-03 and a screening attestation under REG-04 are all instances of the same three-part declaration, which is why an incident reconstruction under REG-05 can consume all four without a translation layer.

Neither the Boundary Invariant nor the BOE Declaration carries a trademark. Both are offered for use, teaching, assessment and extension by anyone, with attribution, under the CC BY 4.0 licence of the specification that names them [2]. Their defensibility rests on a dated, citable priority record rather than on a symbol, and a mark on a proposed law of architecture would suppress the citation the law needs in order to become one.

2.4 Authority flows down, evidence flows up

The composed architecture is read in two directions, and the two directions are not symmetric.

Authority flows down. A binding obligation enters at the Regulatory Floor. The Strategic Compass converts it into a mandate, an appetite and a set of decision rights. Operational Machinery converts the mandate into gates,

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validations, diligence and response procedures. The Technical Substrate converts the procedure into a runtime control. Authority that stops descending at any altitude does not reach the system, and a control the substrate cannot enforce is not a control the institution can rely on.

Evidence flows up. A control writes a BOE Declaration. The gates, validations and attestations aggregate those declarations into gate records. An incident reconstruction consumes them to establish what happened. The maturity profile reports what the aggregate shows. Evidence that stops ascending at any altitude leaves the profile above it asserted rather than tested.

The asymmetry matters and is the registry's central diagnostic claim. Authority can be granted without evidence, which is what a governance programme looks like when it is documented but not enforced. Evidence cannot be produced without authority, because a control that no altitude required is a control no one is accountable for. **An institution MUST therefore be able to trace both directions for any control it operates**, and the inability to trace either is itself a finding.

2.5 Figure 1: the nine as one architecture

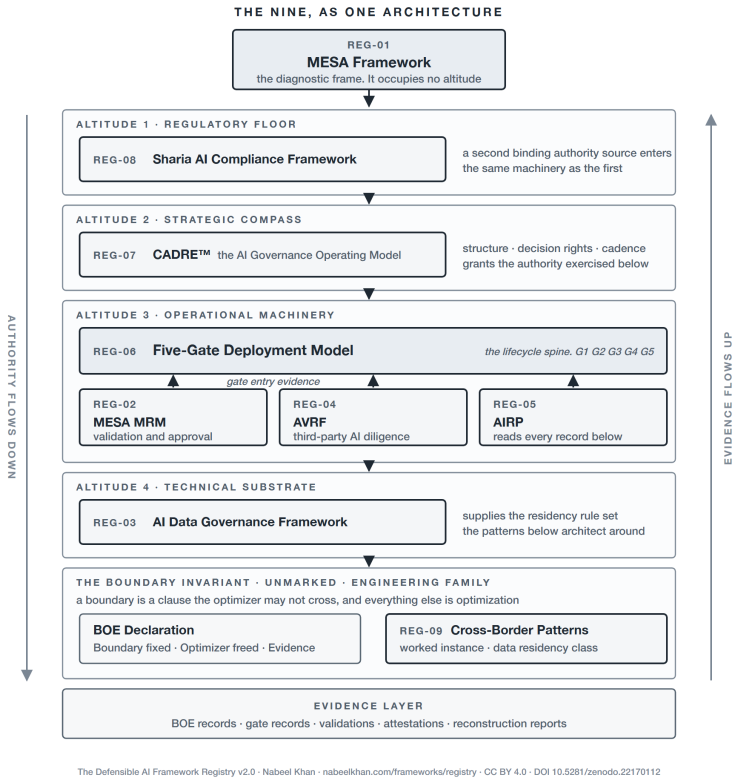


Figure 1. The nine frameworks as one architecture. MESA supplies the four altitudes; each framework instantiates one. CADRE grants authority; the Five-Gate Deployment Model is the lifecycle spine that consumes the outputs of model risk management, vendor risk and data governance as gate entry criteria. The Boundary Invariant governs every control below, and the Cross-Border AI Architecture Patterns are its worked instance on the residency boundary class. Authority flows down; evidence flows up through the BOE Declarations into the evidence layer.

2.6 The two families and the precedence rule

Two families of framework are published by the same author, and they are named and governed differently. Confusing them produces the exact defect this registry was built to remove.

The **governance family** is the nine frameworks catalogued here. Eight of the nine carry trademarks. Their canonical source is this registry.

The **engineering family** consists of A Pattern Language for Production LLM Platforms and the constructs it defines or names: PEVG, PARA, the Boundary Invariant and the BOE Declaration. Every name in that family is deliberately unmarked. Its canonical source is that specification [2], which is itself the standalone restatement of material developed at chapter length across the three volumes of the author's engineering series [14][15][16].

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The precedence rule is normative. Where this registry and the Pattern Language specification describe the same relationship, **the registry governs for the governance family and the Pattern Language specification governs for the engineering family.** A reader encountering a disagreement MUST resolve it in favour of the governing document for the family the disputed construct belongs to, and SHOULD report the disagreement so that the next version of one document or the other removes it.

Stating the precedence is the point. Two documents asserting the same relationships will otherwise drift, and the drift will be discovered by a reader rather than by an author.

2.7 What the architecture does not claim

The architecture claims that the nine compose. It does not claim that an institution must adopt all nine, that adoption in the order given is optimal, or that the decomposition matches the way institutions actually fail. The first is addressed at 8.3, which permits a partial claim. The second is untested and recorded at 10.4. The third is the central untested assertion of MESA itself, which its own specification records, and this registry inherits it rather than resolving it.

---

3. The Registry Entry Schema (Normative)

3.1 The fifteen fields

Every entry in Section 4 carries these fields in this order. An entry that omits a field is not conformant. A field with nothing to record MUST carry an explicit statement that there is nothing to record, because an absent field and an empty field are different claims.

#	Field	What it fixes
1	**Identifier**	The stable REG-nn key. Never renumbered, never reused
2	**Canonical name and mark status**	The name to cite, whether it carries a trademark, and the short form for subsequent mention
3	**Scope phrase**	The generic term the mark modifies at first mention
4	**Specification status**	Whether a standalone specification exists, and its identifier if it does
5	**Definition**	What the framework is, in one paragraph, without reference to its benefits
6	**Purpose**	The question the framework answers for an institution
7	**Altitude or boundary class**	Its position in the architecture of Section 2
8	**Inputs**	What must exist before the framework can operate
9	**Outputs**	What the framework produces that another framework or an authority consumes
10	**Accountable role**	The single named person accountable for its operation
11	**Evidence requirement**	The artifacts that MUST exist for the framework to be considered operating
12	**Relationships**	Its typed edges, per Section 5
13	**Limitations**	What the framework does not do, and what about it is asserted rather than evidenced
14	**Falsification condition**	The observation that would show the framework to be wrong
15	**Change history**	The entry versions in which the entry appeared or changed, each dated

3.2 Specification status, and why versions are not invented

A registry that assigned a version number to every framework would be asserting a version history that does not exist for most of them. This registry does not do that.

Each entry carries a **specification status** drawn from a closed vocabulary of three values.

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- **`deposited`.** A standalone, versioned, citable specification exists and is identified by a DOI. The DOI is the framework's authoritative statement and this entry is a pointer to it.
- **`source-treatment`.** No standalone specification exists. The framework's authoritative treatment is a named chapter of a published work, identified by ISBN and chapter. The entry is the most complete normative statement of the framework available in citable form until a specification is deposited.
- **`instrument-pending`.** No standalone specification exists and one is committed. The entry names what the specification will contain and carries no identifier until it is minted.

A framework's version is the version of its specification, where one exists, and is otherwise undefined. What is versioned in every case is the **entry**, which carries this registry's own version. An entry may change while the framework does not, and frequently will, because a relationship discovered between two frameworks changes both entries and neither framework.

3.3 The accountable role is a person

Every entry MUST name a single accountable role, and that role MUST be occupiable by one person. A committee, a function, a team or a board MUST NOT be recorded as the accountable role.

This is the property that distinguishes a governance framework from a process description, and it is stated here once rather than nine times. A committee can be consulted, can review, can challenge and can decide. It cannot be accountable, because accountability that is distributed across a body is accountability no member of that body carries alone, and the question an authority asks after a failure is which person is answerable.

Where a body holds binding approval authority, as a Sharia Supervisory Board does under REG-08, the body is recorded as an **authority source** in the inputs and the named institutional officer who carries its determinations into the lifecycle is recorded as the accountable role. The two are different fields because they are different obligations.

Where a framework operates at more than one locus, the entry names one accountable role per locus, each occupiable by one person, and records the loci. Three entries do so and each states why: REG-02 separates the discipline from the individual approval, because the independence of the validation is the property the framework exists to establish; REG-03 assigns per data domain, because one accountable person for all institutional data is a person accountable for nothing specific; and REG-06 assigns per gate, because the gate is the locus. Plurality of locus is not plurality of accountability. **What remains forbidden in every case is a role no single person can occupy.**

3.4 The evidence requirement is a test, not an aspiration

An entry's evidence requirement MUST name artifacts that either exist or do not. **A framework whose evidence artifacts are not produced is not operating**, however completely it is documented, and an entry MUST NOT record intent, policy or training as evidence of operation.

This rule is inherited rather than invented. It is the same rule the MESA Framework applies through its enforcement constraint, under which a control the technical substrate cannot enforce operates at the maturity of the substrate rather than at the maturity of the policy, and the same rule the Pattern Language applies through its third scoring rule, under which a pattern whose evidence artifact is not produced is not implemented.

3.5 Limitations are stated by the entry, not inferred from it

Every limitation carries one of four kinds, and the kind is the first word of the entry that states it, so that the prose and the machine-readable layer can be compared without interpretation.

Kind	What it records
Asserted	A claim made in the framework's source treatment that has not been independently checked
Missing	A named artifact the framework needs and does not have
Scope	What the framework does not govern, stated so that a reader does not expect it to
Normative disclosure	A clause that MUST accompany any presentation of the framework, because omitting it would misrepresent what the framework is

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Every entry MUST record both of the following where either applies.

- **What is asserted rather than evidenced.** A claim made in the framework's source treatment that has not been independently checked. Marking it does not weaken the framework. It tells a reader which claims to verify before relying on them, and an entry that marks none is claiming a completeness no framework in this registry has.
- **What is missing and known to be missing.** A named artifact the framework needs and does not have.

3.6 The falsification condition

Every entry MUST state one observation that would show the framework to be wrong. **A restatement of the framework's limitations is not a falsification condition**, and neither is an observation that the framework was applied badly. The condition MUST be capable of being observed by a party unconnected to the author, and MUST bear on the framework's central claim rather than on its adoption.

3.7 Change history

Every entry MUST carry a change history recording, at minimum, the entry version in which it first appeared. Renames, consolidations, supersessions and repositionings MUST each appear as a dated line, because the registry is the citable instrument of those decisions and an undated decision is not evidence of anything.

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4. The Nine Registry Entries (Normative)

Each entry below is the authoritative statement of what its framework is called, what it requires, what it produces and how it relates to the others. Entries are ordered by identifier. The identifier, not the position, is the citation key.

4.1 REG-01 · MESA Framework

4.1.1 Identifier. REG-01.

4.1.2 Canonical name and mark status. The **MESA Framework™**. Trademarked. Short form after first mention: **MESA**.

MESA stands for **Maturity, Evidence, Substrate, Alignment**. Each term is drawn from language already present in the framework's specification rather than fitted to the letters: the framework reports a maturity vector, each altitude carries an evidence requirement an auditor can test, the Technical Substrate bounds every runtime-dependent control above it, and institutional AI governance is an alignment condition across altitudes. **The four terms are not the four altitudes.** Substrate appears in both lists because the Technical Substrate is both an altitude and the property that makes the enforcement constraint possible, and no other correspondence between the two lists is intended.

Through version 1.0 of the specification and in the source book, MESA was expanded as *Middle East Strategic Alignment*. That expansion is retired as of specification version 1.1, recorded at Appendix A.5 and in this entry's change history, because it described the framework's origin rather than its scope. The mark, the four altitudes, the five mechanisms and the conformance language are unchanged.

4.1.3 Scope phrase. A four-altitude diagnostic model for institutional AI governance.

4.1.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22109836 [1]. The DOI given is the concept DOI and always resolves to the latest version.

4.1.5 Definition. MESA decomposes institutional AI governance into four altitudes, each carrying a definition, its inputs and outputs, one accountable role, and an evidence requirement an auditor can test. The altitudes are the **Regulatory Floor**, which carries what binds the institution from outside; the **Strategic Compass**, which carries what the institution chooses; **Operational Machinery**, which carries what the governance function performs; and the **Technical Substrate**, which carries what the system enforces at runtime. Five mechanisms operate the framework: layer separation, the cross-layer cascade, the enforcement constraint, mandatory interface couplings, and the evidence-based maturity vector.

doi.org/10.5281/zenodo.22170112

4.1.6 Purpose. To answer the question a board or a supervisor asks as a single question and which cannot be answered as one: how mature is our AI governance. MESA answers it as a profile that names which altitude is failing, on the ground that a single grade is precisely what allows a strong regulatory posture to conceal a substrate that cannot support it.

4.1.7 Altitude. None. MESA is the frame in which the altitudes are defined and does not occupy one. This is recorded as a property of the entry rather than as a missing field.

4.1.8 Inputs. The institution's governance artifacts: its obligation register, charter, risk appetite statement and decision rights. The system evidence produced by the frameworks at each altitude. The scope of the AI estate being assessed.

4.1.9 Outputs. A per-altitude maturity vector, for example a four-position profile in which the fourth position reports the Technical Substrate independently of the three above it. The result of the enforcement-constraint test, which states the effective rather than the documented state of every runtime-dependent control.

4.1.10 Accountable role. The single executive accountable for the institution's AI governance mandate, who is the recipient of the profile and the person answerable for the gap it names. The four altitude-level accountable roles are specified in the framework specification and are not restated here.

4.1.11 Evidence requirement. A maturity assessment is not operating unless all of the following exist: the obligation register and its mapping to owners; the chartered mandate and documented decision rights; gate records and completed validations; and runtime policy configuration, event logs and BOE records from the substrate. **The fourth of these is the one whose absence caps the result**, and an assessment that reports a vector without substrate evidence is reporting a documented posture rather than an enforced one.

4.1.12 Relationships. MESA is the parent frame of REG-02 through REG-08, each of which instantiates exactly one altitude. It absorbs the Governance Maturity Model, whose supersession is recorded at 6.3. It has no relationship of dependency to REG-09, which sits under the Boundary Invariant rather than under an altitude. Full edges at 5.2.

4.1.13 Limitations.

- *Asserted.* That the four-altitude decomposition matches the way institutions actually fail. The specification records this as untested and this registry inherits the assertion without resolving it.
- *Asserted.* The clause-level mappings to ISO/IEC 42001 [17] and the NIST AI Risk Management Framework [18] are the author's readings of those documents. Neither issuing body has reviewed or endorsed them.
- *Missing.* No scoring dataset has been published, so the distribution of assessed profiles across institutions is unknown and no baseline exists against which a result can be positioned.
- *Missing.* No institution unconnected to the author has published a MESA assessment.

4.1.14 Falsification condition. MESA is falsified if institutions that score high at the Regulatory Floor, the Strategic Compass and Operational Machinery while scoring low at the Technical Substrate are found to experience AI governance failures at the same rate as institutions scoring high at all four. That finding would refute the enforcement constraint, which is the mechanism the four-altitude decomposition exists to express, and a framework whose central mechanism does not hold does not survive by being useful.

4.1.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
1.0	2026-08-30	Records the retirement of the expansion *Middle East Strategic Alignment* at specification version 1.1, and the absorption of the Governance Maturity Model

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4.2 REG-02 · MESA MRM Framework

4.2.1 Identifier. REG-02.

4.2.2 Canonical name and mark status. The **MESA MRM Framework™**. Trademarked. Short form after first mention: **MESA MRM**.

doi.org/10.5281/zenodo.22170112

4.2.3 Scope phrase. A six-step model risk management discipline.

4.2.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22285045 [8]. The DOI given is the concept DOI and always resolves to the latest version. Deposited on 15 September 2026, and **this entry is reconciled with that specification at this version, the specification governing under 11.3**. The source treatment remains chapter 12 of *AI Governance and Compliance Frameworks for the Middle East: The Enterprise Playbook* [13].

4.2.5 Definition. MESA MRM specifies model risk management as an institutional discipline in six steps: **inventory**, in which every model is registered, given a named owner and assigned a risk tier that determines the depth of everything following; **pre-validation**, in which data lineage, assumptions and stated limits are established by the first line; **independent validation**, performed by a second line with no stake in the model passing; **approval**, in which a named accountable person signs on the evidence rather than on the recommendation; **monitoring**, which tracks drift, override rate and the conditions the approval was conditioned on; and **revalidation**, triggered by monitoring rather than by the calendar alone.

The framework rests on one principle, stated here in the operative form the specification gives it [8]: **a validation is not a property of the model but a bounded permission, held by the institution, over a stated use, for a stated period.** A property would travel with the object, and a permission does not: a use outside what was named is unpermitted whether or not anyone noticed. Version 1.0 of this registry stated the same principle in its shorter form, that validation confers the right to be believed. That form is not withdrawn and remains the one to quote to a reader; the operative form above is the one an artifact can be tested against. A model that has not been independently validated may be operated, but its outputs are not evidence of anything to a party outside the team that built it.

Where a Sharia Supervisory Board holds binding approval power, step three carries a **dual-validation branch**: model logic passes a civil validation track and a Sharia validation track as two independent tracks, and both MUST close before approval. The branch is jointly specified with REG-08.

4.2.6 Purpose. To make a model's fitness a matter of record rather than a matter of the building team's confidence, and to make the point at which it becomes a matter of record identifiable in time.

4.2.7 Altitude. Operational Machinery.

4.2.8 Inputs. The model inventory and its tiering rubric. Data lineage and classification records from REG-03. Vendor evidence from REG-04 where the model is third-party. Revalidation triggers from REG-05, which reopen the cycle at step six. The Sharia Supervisory Board's determinations where REG-08 applies. The approval authority granted by REG-07.

4.2.9 Outputs. Validation reports. Approval records naming the person, the evidence and the date. Monitoring findings. Revalidation triggers. These are consumed as entry criteria at gates two and three of REG-06 and contribute to the Operational Machinery position of the REG-01 maturity vector.

4.2.10 Accountable role. The head of model risk, in the second line, is accountable for the discipline. The named approving executive is accountable for each individual approval. These are two roles and MUST NOT be held by one person for the same model, because the independence of the validation is the property the framework exists to establish.

4.2.11 Evidence requirement. A model inventory with an owner and tier per entry. One validation report per model at or above the tier threshold. One approval record per production model, naming a person. Monitoring output covering the period since approval. A revalidation trigger register. **A validation report that no second-line reviewer signed is not evidence of independent validation**, and an approval record naming a committee rather than a person does not satisfy 3.3.

4.2.12 Relationships. Child of REG-01 at Operational Machinery. Supplies evidence to REG-06 at gates two and three, and the revalidation trigger register at gate five. **The gate five supply was added at version 2.0**, resolving the disagreement the specification records at its 6.3 and routes here: Five-Gate names that register as a gate five entry criterion and attributes it to this framework [3, 3.6], and version 1.0 of this register carried the edge at gates two and three only. The register was incomplete. Consumes classification and lineage from REG-03, vendor evidence from REG-04, and revalidation triggers from REG-05. Threaded by REG-08 through the dual-validation branch. Operates under authority granted by REG-07. Its records are consumed by REG-05 during reconstruction. Full edges at 5.2.

4.2.13 Limitations.

doi.org/10.5281/zenodo.22170112

- *Asserted.* Alignment with supervisory model risk expectations. The source treatment does not contain a written crosswalk to the current Canadian guideline on model risk or to United States supervisory guidance. An institution relying on this framework for supervisory alignment MUST perform that crosswalk against the primary sources in force in its jurisdiction, and this registry does not perform it. The deposited specification carries an **informative** crosswalk at its Annex A, which states its own date of reading, creates no obligation on either supervisor and no presumption in the institution's favour, and expressly leaves the obligation where it was [8]. **The limitation therefore stands and is amended rather than closed at this version.**
- *Scope.* The framework specifies the discipline, not the statistical or behavioural methods a validator applies within it. It does not tell a validator how to validate.

4.2.14 Falsification condition. MESA MRM is falsified if institutions operating the six steps with genuine second-line independence are found to approve unfit models at the same rate as institutions in which the building team validates its own work. The framework's entire claim rests on independence changing the outcome, and evidence that it does not would leave a process with no defensible reason to exist.

4.2.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
2.0	2026-09-15	Status moves to `deposited` on the deposit of the MESA MRM Framework specification [8], whose concept DOI the entry now carries
2.0	2026-09-15	Definition reconciled with that specification, which states the principle of 4.2.5 in its operative form and governs
2.0	2026-09-15	The `missing` limitation recording no validation report template and no worked example is closed by Annexes B and C of that specification
2.0	2026-09-15	The asserted supervisory-alignment limitation is amended to record the informative crosswalk at its Annex A, and is NOT closed by it
2.0	2026-09-15	The supply edge to REG-06 gains the revalidation trigger register at gate five, resolving the question routed here at 6.3 of that specification

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4.3 REG-03 · AI Data Governance Framework

4.3.1 Identifier. REG-03.

4.3.2 Canonical name and mark status. The **AI Data Governance Framework™**. Trademarked. Short form after first mention: **AI Data Governance**.

Renamed. This framework was asserted in the source book as the *AI Data Governance Stack*. The rename is recorded at 6.4 and in this entry's change history. The former name remains valid for citation of the earlier work and MUST NOT be treated as an error in it.

4.3.3 Scope phrase. A control system for the data boundary in AI systems.

4.3.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22285047 [9]. The DOI given is the concept DOI and always resolves to the latest version. Deposited on 14 September 2026, and **this entry is reconciled with that specification at this version, the specification governing under 11.3**. The source treatment remains chapter 13 of the Enterprise Playbook [13].

4.3.5 Definition. AI Data Governance specifies the control system for the data boundary: what data may exist inside an AI system, where it may reside, how its lineage is proven, what quality and minimization gates it passes, and what certification it carries. It operates in five stages. **Classify** assigns sensitivity, personal-data status, residency class and, where REG-08 applies, permissibility. **Bound** fixes the residency rule set, which is the set of clauses that MUST NOT be crossed and which is supplied onward to REG-09. **Prove** constructs the lineage graph from source through each transformation to training or inference use, with each hop attested. **Gate** applies quality thresholds, the minimization check and certification. **Release** admits the data to model or agent use with a BOE record attached.

The framework carries the **Halal data certification chain**, a source-to-use permissibility chain specified with REG-08 for institutions under Sharia authority.

doi.org/10.5281/zenodo.22170112

4.3.6 Purpose. To make the question of whether a given item of data may be in a given AI system answerable from a record rather than from a recollection, and to make the answer survive the departure of the person who gave it.

4.3.7 Altitude. Technical Substrate. In the engineering family, this framework is the worked treatment of the **data boundary class** defined in the Pattern Language specification [2].

4.3.8 Inputs. The data inventory. The applicable residency and personal-data obligations from the Regulatory Floor. The classification scheme in force. Sharia permissibility criteria where REG-08 applies.

4.3.9 Outputs. Classification records. Residency attestations. Lineage graphs. Certification records including, where applicable, the Halal data certification chain. The residency rule set consumed by REG-09. All of these are consumed as entry criteria at gate one of REG-06.

4.3.10 Accountable role. The named data owner for the data domain. Accountability is per domain rather than per institution, because a single accountable person for all institutional data is a person accountable for nothing specific.

4.3.11 Evidence requirement. A classification record per data set admitted to an AI system. A residency attestation per jurisdiction in which data is held or processed. A lineage graph reaching from an attested source to the point of use. Certification records where a certification is claimed. **A lineage graph with an unattested hop is not a lineage graph**, because the purpose of the artifact is to be reconstructable end to end and a single gap defeats the reconstruction.

4.3.12 Relationships. Child of REG-01 at the Technical Substrate. Supplies the residency rule set to REG-09, which architects around it. Supplies evidence to REG-06 at gate one and to REG-02 at pre-validation. Threaded by REG-08 through the Halal data certification chain. Its records are consumed by REG-05 during reconstruction. Full edges at 5.2.

4.3.13 Limitations.

- *Normative disclosure.* The Halal data certification chain is an original construct of this framework. **No standard-setter recognizes such a chain as of the date of this registry.** It MUST be presented as author methodology and MUST NOT be presented as an industry or supervisory requirement.
- *Scope.* The framework governs the data boundary. It does not specify data quality measurement methods, storage architecture or transformation tooling, and it is not a replacement for a general data management body of knowledge. It extends one.

4.3.14 Falsification condition. AI Data Governance is falsified if institutions maintaining attested lineage to the point of use are found unable to answer a data provenance question from an authority any faster, or any more defensibly, than institutions maintaining classification alone. The framework's cost sits almost entirely in the attestation of each hop, and evidence that the attestation adds nothing to the answer would remove its justification.

4.3.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
1.0	2026-08-30	Records the rename from *AI Data Governance Stack*, and the repositioning as the worked treatment of the data boundary class
2.0	2026-09-15	Status moves to `deposited` on the deposit of the AI Data Governance Framework specification [9], whose concept DOI the entry now carries
2.0	2026-09-15	Definition reconciled with that specification, to record the residency rule set and the JSON Schema published with it
2.0	2026-09-15	The `missing` limitation recording no machine-readable classification schema is **closed** by `residency-rule-set-schema-v1.0.json`, deposited with it

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4.4 REG-04 · AI Vendor Risk Framework (AVRF)

4.4.1 Identifier. REG-04.

doi.org/10.5281/zenodo.22170112

4.4.2 Canonical name and mark status. The **AI Vendor Risk Framework (AVRF)™**. Trademarked. Short form after first mention: **AVRF**.

4.4.3 Scope phrase. A due-diligence discipline for third-party AI.

4.4.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22170146 [7]. The DOI given is the concept DOI and always resolves to the latest version. The authoritative treatment of the wider framework remains chapter 14 of the Enterprise Playbook [13]. This entry is updated in the registry version following that deposit.

4.4.5 Definition. AVRF specifies the governance of AI risk that enters the institution through a third party. It operates in five stages. **Classify** places the vendor on a criticality tier determined by three factors together: business criticality, the institution's data exposure to the vendor, and the autonomy the vendor's AI is permitted over institutional decisions. **Diligence** applies a structured questionnaire covering provenance, training data, evaluation results, incident history, subprocessors, model-change notification, and exit and portability, with an answer format and an evidence expectation per question so that answers are comparable across vendors. **Contract** fixes audit rights, evaluation rights, incident notice, model-change notice, and exit and portability as terms rather than as expectations. **Monitor** tracks drift on the vendor side, version changes, and concentration across the vendor estate. **Exit** requires that a substitution plan has been tested, not merely written.

Where REG-08 applies, the diligence stage carries a **vendor screening branch** applying the Sharia Supervisory Board's criteria to the vendor's model conduct.

4.4.6 Purpose. To make third-party AI risk assessable on the same terms as first-party AI risk, across a vendor boundary the institution cannot inspect directly, and which may carry a large share of its AI exposure.

4.4.7 Altitude. Operational Machinery.

4.4.8 Inputs. The vendor inventory and the criticality tiering rubric. The data classification records from REG-03 that establish exposure. The contractual position in force. Sharia screening criteria where REG-08 applies.

4.4.9 Outputs. Vendor risk records. Completed diligence questionnaires with the evidence supplied against each answer. Contractual control registers. Monitoring logs and concentration analysis. Screening attestations where applicable. Tested exit plans. These are consumed as entry criteria at gates one and four of REG-06.

4.4.10 Accountable role. The named owner of the vendor relationship. Where a vendor is used by more than one business line, one owner is named for the relationship and the consuming lines are recorded as consulted, because two owners of one vendor is the condition under which neither monitors it.

4.4.11 Evidence requirement. A vendor risk record per vendor supplying AI capability. A completed questionnaire with evidence attached, dated and versioned to the questionnaire version used. A contractual control register naming which of the five contract terms were obtained and which were declined. Monitoring output covering the period since onboarding. A dated record of the most recent exit test. **A questionnaire answered without evidence attached is an assertion by the vendor and MUST be recorded as such**, and an exit plan that has never been tested MUST NOT be recorded as an exit capability.

4.4.12 Relationships. Child of REG-01 at Operational Machinery. Supplies evidence to REG-06 at gates one and four, and to REG-02 where the model under validation is third-party. Consumes exposure classification from REG-03. Threaded by REG-08 through vendor screening. Its records are consumed by REG-05 during reconstruction. Full edges at 5.2.

4.4.13 Limitations.

- *Asserted.* That the questionnaire structure matches supervisory expectations for third-party model risk. No supervisor has reviewed the instrument.
- *Missing.* The questionnaire is now deposited, so the instrument exists. What remains missing is evidence that two institutions applying it independently produce comparable answers, which is a property use confers rather than publication.
- *Scope.* AVRF governs the assessment of the vendor. It does not govern the selection decision, the commercial negotiation, or the security assessment of the vendor's infrastructure, which are separate disciplines the framework assumes are performed elsewhere.

4.4.14 Falsification condition. AVRF is falsified if vendors assessed under the full five stages are found to produce AI incidents in the acquiring institution at the same rate and severity as vendors assessed under a generic

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third-party security questionnaire. The framework's claim is that model-specific diligence changes the outcome, and evidence that it does not would make the additional stages ceremony.

4.4.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry, status `deposited`

4.5 REG-05 · AI Incident Response Protocol (AIRP)

4.5.1 Identifier. REG-05.

4.5.2 Canonical name and mark status. The **AI Incident Response Protocol (AIRP)™**. Trademarked. Short form after first mention: **AIRP**.

4.5.3 Scope phrase. An AI incident response and reconstruction protocol.

4.5.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22285057 [10]. The DOI given is the concept DOI and always resolves to the latest version. Deposited on 6 September 2026, and **this entry is reconciled with that specification at this version, the specification governing under 11.3**. The source treatment remains chapter 15 of the Enterprise Playbook [13].

4.5.5 Definition. AIRP specifies what an institution does when an AI system fails, in six stages. **Signal** admits a detection trigger: a drift alarm, an override spike, a complaint, a gate breach or an external report. **Classify** assigns severity, determines regulatory notifiability, and where REG-08 applies determines Sharia materiality. **Contain** applies rollback, routes around the failing component, or suspends the agent authority in question. **Escalate** routes along the paths REG-07 defines, to the governance office, the executive, the board, the regulator or the Sharia Supervisory Board as the classification requires. **Reconstruct** establishes what happened from the BOE records the other frameworks produced: which boundary was in force, which policy version applied, which evidence existed at the moment of the decision, and which person approved it. **Close** produces a postmortem, a revalidation trigger into REG-02, and a conformance test that would have caught the failure.

The protocol's hardest requirement is stated in its fifth stage rather than its first. **An incident is explainable only if the evidence needed to reconstruct it existed at the moment of the decision, bound to the policy version then in force.** Evidence assembled after the fact is a reconstruction of the institution's beliefs rather than of the system's behaviour.

4.5.6 Purpose. To make an AI failure answerable, which is a different requirement from making it survivable. An institution can contain an incident it cannot explain, and it is the explanation an authority asks for.

4.5.7 Altitude. Operational Machinery.

4.5.8 Inputs. Detection triggers armed at gate five of REG-06. The BOE records produced by every other framework in this registry except REG-01, which supplies evidence to nothing. The escalation paths and decision rights defined by REG-07. Notification obligations from the Regulatory Floor. Sharia materiality criteria where REG-08 applies.

4.5.9 Outputs. Incident records. Reconstruction reports. Regulator notifications where required. Revalidation triggers into REG-02. Conformance tests derived from the failure.

4.5.10 Accountable role. The named incident lead in the governance office. The role is named in advance and MUST be occupiable at the time an incident occurs, which means a named deputy is part of the appointment rather than an addition to it.

4.5.11 Evidence requirement. An incident record per admitted signal, including those closed as non-incidents, because a signal register that records only confirmed incidents cannot show the detection rate. A reconstruction report for every incident classified as notifiable. A dated notification record where a notification was made or a dated rationale where one was considered and not made. A closure record carrying the revalidation trigger and the derived conformance test.

4.5.12 Relationships. Child of REG-01 at Operational Machinery. **Consumes the evidence of every other framework in this registry except REG-01**, which is the property that makes the evidence layer non-optional

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rather than merely recommended. Loops back into REG-02 through revalidation and into REG-06 by reopening gate two. Escalates along paths defined by REG-07. Full edges at 5.2.

4.5.13 Limitations.

- *Asserted, and jurisdiction-dependent.* Notification timelines and the threshold at which an incident becomes notifiable vary by jurisdiction and by authority. The source treatment states them for its calibrating jurisdictions. **An institution MUST re-verify both against the primary sources in force in its own jurisdiction**, and this registry does not state them.
- *Scope, and a dependency stated as a limitation because it is one.* Reconstruction is possible only to the extent that the other frameworks kept their evidence. AIRP cannot compensate for an upstream framework that produced none, and an institution operating AIRP over frameworks that write no BOE records has an escalation procedure rather than a reconstruction capability.

4.5.14 Falsification condition. AIRP is falsified if institutions holding complete upstream BOE records are found no better able to reconstruct an AI incident, to a standard an external authority accepts, than institutions holding conventional application and infrastructure logs. The protocol's distinguishing claim is that decision-time evidence bound to a policy version is what reconstruction requires, and evidence that ordinary logging suffices would collapse the distinction the framework is built on.

4.5.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
1.0	2026-08-30	Records the reframing of the protocol around reconstruction rather than crisis process
2.0	2026-09-15	Status moves to `deposited` on the deposit of the AI Incident Response Protocol specification [10], whose concept DOI the entry now carries
2.0	2026-09-15	The `missing` limitation recording no incident taxonomy is **closed** by `airp-incident-taxonomy-v1.0.json`, deposited with it and aligned to the two public repositories at [26] and [27]

4.6 REG-06 · Five-Gate Deployment Model

4.6.1 Identifier. REG-06.

4.6.2 Canonical name and mark status. The **Five-Gate Deployment Model™**, a deployment discipline for AI systems. Trademarked. Short form after first mention: **Five-Gate**.

Naming and scope. The model governs the passage of a model, agent or AI-enabled service from design into production and through to retirement, under institutional accountability and evidence requirements. It is unrelated to, and not derived from, stage-gate processes for new product development, which denote an idea-to-launch process for product innovation: a different subject in a different field. No affiliation, endorsement or lineage is claimed or implied. The terms *stage-gate*, *phase-gate* and *gating process* MUST NOT be used as synonyms for Five-Gate on any surface, and this rule applies to the registry, the specification, the source book and any derived material.

4.6.3 Scope phrase. A deployment discipline for AI systems.

4.6.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22170122 [3]. The DOI given is the concept DOI and always resolves to the latest version. The source treatment is chapter 10 of the Enterprise Playbook [13], with the gate structure appearing across the lifecycle chapters. This entry is updated in the registry version following that deposit.

4.6.5 Definition. Five-Gate is the lifecycle spine of the governance family. **No AI system reaches production, or remains there, except by passing five gates in order.** Each gate carries entry criteria supplied by another framework in this registry, one accountable owner, and a required evidence record.

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Gate	Name	Entry evidence supplied by	Accountable role
G1	Data and Design	REG-03 classification and lineage; REG-04 vendor clearance where the component is third-party	The named data owner
G2	Validation	REG-02 steps one and two	The named validator
G3	Approval	REG-02 steps three and four; REG-08 dual validation closed where applicable	The named accountable executive
G4	Deployment	REG-07 release authority; REG-04 contractual controls in force; REG-09 deployment topology where the system spans jurisdictions; guardrails configured	The named platform owner
G5	Operation	REG-05 triggers armed; monitoring live	The named head of the governance office

The sequence and the entry criteria are normative. A gate passed without its entry evidence is not passed, and a deployment that skipped a gate is not a deployment the institution can defend.

Two paths run backwards through the model and are part of it rather than exceptions to it. **Rollback** reverses G5 to G4 with the evidence of the reversal preserved. **Loop-back** reopens G2 when an incident occurs at G5, so that a system that failed in operation re-enters validation rather than being patched in place.

4.6.6 Purpose. To make the question a supervisor or an auditor asks in one sentence answerable in one document: show me how a model gets to production here, and who said it could.

4.6.7 Altitude. Operational Machinery. Five-Gate is the horizontal thread through that altitude and has the largest set of upstream suppliers in the registry: six of the eight other entries supply evidence it requires before a gate can be passed.

4.6.8 Inputs. The outputs of REG-02, REG-03 and REG-04 as gate entry criteria. Release authority from REG-07. Detection triggers from REG-05 for gate five. The deployment topology from REG-09 where the system spans jurisdictions. Dual-validation closure from REG-08 where it applies.

4.6.9 Outputs. One gate passage record per gate, naming who approved, on what evidence, and when. Rollback records where a reversal occurred. Loop-back records where an incident reopened validation. These records are the institution's primary evidence at the Operational Machinery position of the REG-01 maturity vector.

4.6.10 Accountable role. One named person per gate, as listed at 4.6.5. **A committee MUST NOT be recorded as the accountable role at any gate.** This is the property that distinguishes Five-Gate from a delivery pipeline: a pipeline enforces a sequence, and a gate assigns a person to it.

4.6.11 Evidence requirement. Five gate passage records per production system, each naming a person and citing the entry evidence relied on. **Each gate passage record is an instance of the BOE Declaration**: the gate's entry criteria are the boundary it fixes, the deployment freedom granted on passage is the optimizer it frees, and the record itself is the evidence that the boundary held. This mapping is what makes Five-Gate compose with the engineering family rather than merely coexist with it.

4.6.12 Relationships. Child of REG-01 at Operational Machinery. Consumes evidence from REG-02, REG-03 and REG-04, the dual-validation closure of REG-08 at gate three, and the deployment topology of REG-09 at gate four. Operates under authority granted by REG-07. Consumes the revalidation trigger register of REG-02 at gate five, which 4.6.5 names as an entry criterion there and which the register carries from version 2.0. Reopened by REG-05 through loop-back. Approves the topology produced by REG-09. Its gate records are consumed by REG-05 during reconstruction. Full edges at 5.2, and the composition is drawn at Figure 2.

4.6.13 Limitations.

- *Asserted, and a name risk. It is the reason for the naming rule at 4.6.2.* The general concept of gated progression is dominated in an adjacent field by a registered mark held in the product-innovation domain. Five-Gate is verbally distinct and operates in a different field. **A trademark clearance search has not been completed as of the date of this registry**, and the naming rule at 4.6.2 is stated as a discipline rather than as the outcome of one. This registry does not assert that the mark is available.
- *Missing.* No readiness assessment instrument exists, so an institution can be told what the gates require but cannot be scored against them.

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- *Missing.* No gate evidence templates are published, so what a gate three sign-off looks like is left to each institution to invent.
- *Scope.* Five-Gate governs passage. It does not specify what a validation must contain, what a classification must cover, or what a monitoring threshold should be. Those are the frameworks it consumes, and a gate is only as strong as the evidence handed to it.

4.6.14 Falsification condition. Five-Gate is falsified if institutions operating five gates with a single named accountable person at each are found to deploy unfit AI systems at the same rate as institutions operating an automated delivery pipeline with equivalent automated checks and no named approver. The model's distinguishing claim is that a named person accountable at a checkpoint changes the outcome relative to an equivalent automated check, and evidence that it does not would leave the gates as latency.

4.6.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry, status `deposited`
1.0	2026-08-30	Records the promotion from lifecycle component to the lifecycle spine of the governance family, and the BOE mapping at 4.6.11
2.0	2026-09-15	The inbound supply edge from REG-02 gains the revalidation trigger register at gate five, which this framework already named as an entry criterion. The entry changes; the framework does not

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4.7 REG-07 · CADRE

4.7.1 Identifier. REG-07.

4.7.2 Canonical name and mark status. CADRE™, the AI Governance Operating Model. **Trademarked** from 1 September 2026. Short form after first mention: **CADRE**. The identifier REG-07 is unchanged by the rename and MUST NOT change at any future one. **The former canonical name is retained as the framework's descriptor** and is not withdrawn from use: the canonical first mention pairs the two, and the short form recorded at version 1.0, *the Operating Model*, is replaced by *CADRE*. Recorded at 6.9 and Appendix A.2.

4.7.3 Scope phrase. The structure, decision rights and cadence of the AI governance function, specified as five artifacts an auditor can test.

4.7.4 Why this framework is marked. Version 1.0 recorded this entry as deliberately unmarked and recorded one question as open: whether to coin a distinctive name for the consolidated framework, keeping the descriptive phrase as its descriptor. **That question is now closed in the affirmative**, and the reasoning that kept the former name unmarked is rewritten here rather than deleted, because the record of why it was open is part of the answer.

The reasoning was that *AI Governance Operating Model* is a descriptive phrase in common industry use; that asserting a mark on it would be hard to register, weak to enforce, and would invite exactly the criticism the consolidation recorded at 6.2 exists to remove; and that the value of this framework sits in its artifacts rather than in the exclusivity of its name. **None of that is withdrawn.** The rename answers it rather than overruling it: a distinctive name is coined, the descriptive phrase is kept as the descriptor, and the artifacts are unchanged.

NO TRADEMARK CLEARANCE SEARCH HAS BEEN COMPLETED for the name CADRE. No registration has issued, no registered-mark symbol is used, and this registry does not assert that the mark is available, registrable or free of conflict. That is an open limitation and is recorded at 10.6 alongside the two frameworks already carrying it. The derivation of the name, and the full record of the decision, are at Appendix A of the deposited specification [11] and are carried here by reference rather than restated.

4.7.5 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22285052 [11]. The DOI given is the concept DOI and always resolves to the latest version. Deposited on 8 September 2026 **under the name CADRE**, one week before this registry version carried the rename; a reader who followed the DOI from version 1.0 of this registry therefore reached a record whose title did not match the entry pointing at it, which that specification anticipates at its B.4. **This entry is reconciled with that specification at this version, the specification governing under 11.3**. The source treatment remains chapters 10 and 11 of the Enterprise Playbook [13], read together with the consolidation recorded at 6.2.

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4.7.6 Definition. CADRE specifies how the governance function itself exists, in three parts that were formerly three separate frameworks.

- **Structure.** The governance office: its roles, its staffing, its reporting lines, and its position within the three-lines model in which the first line owns the risk, the second line challenges it, and the third line audits both. The named roles include the head of the office and leads for model risk, data, vendor and incident, together with a Sharia liaison where REG-08 applies.
- **Decision rights.** The assignment, per lifecycle phase and per decision type, of who is accountable, who is responsible, who is consulted and who is informed. **Exactly one person is accountable per decision.** The gate three approval of REG-06 is an assignment made here rather than a separate authority.
- **Cadence.** The rhythm on which the function operates: a weekly operations review, a monthly risk review, a quarterly board report and an annual charter refresh, together with the escalation paths on which incidents, human-review tiers and gate disputes terminate.

The framework also specifies a ninety-day stand-up path from board mandate through charter and office to first operating cadence.

The deposited specification states these same three parts as **five artifacts an auditor can test**: Charter, Authority, Decision rights, Records and Escalation [11]. The five are what the three parts produce and are not a restatement of them. Structure produces the Charter and the Authority; Cadence produces the Records and the Escalation; Decision rights is both a part of this definition and an artifact, which is why it appears in both lists. **No other correspondence between the three and the five is intended**, and the three parts above remain the definition.

4.7.7 Purpose. To convert a mandate into authority that is exercised on a schedule and recorded, which is the difference between a governance function and a governance intention.

4.7.8 Altitude. Strategic Compass.

4.7.9 Inputs. The board mandate. The institution's risk appetite. The regulatory obligations that constrain the mandate. The three-lines structure already in force for non-AI risk, which the AI function extends rather than duplicates.

4.7.10 Outputs. The charter. The decision-rights assignment. The cadence calendar and the meeting and escalation records it produces. Release authority consumed at gate four of REG-06. The escalation paths consumed by REG-05.

4.7.11 Accountable role. The named head of the AI governance office.

4.7.12 Evidence requirement. A charter, dated and approved by the body that granted the mandate. A decision-rights assignment naming one accountable person per decision type. Records of each cadence meeting held, including those cancelled and the reason. Escalation records. **A cadence with no meeting records is not a cadence**, and a decision-rights assignment naming a committee as accountable does not satisfy 3.3.

4.7.13 Relationships. Child of REG-01 at the Strategic Compass. Grants authority to REG-02, REG-03, REG-04, REG-05, REG-06 and, from version 2.0, REG-09, and supplies to REG-05 the escalation paths along which incidents route. **The grant to REG-09 was added at this version**, resolving the gap that framework's specification records at its 8.2 and routes here: the register granted authority to five frameworks and none to REG-09, while REG-09 produces evidence at rows 26 and 27, which 2.4 holds cannot happen without authority. The gap was in this register. A reader will find [11] stating that this framework grants authority to five others; that was true of the register published at version 1.0 and is superseded here under 11.3. The register carries no receipt edge, because what returns to this framework is an escalation record it writes itself rather than an artifact another framework produces. Absorbs three frameworks formerly asserted separately, recorded at 6.2. Full edges at 5.2.

4.7.14 Limitations.

- *Asserted.* That the structure fits institutions of different sizes. The source treatment is calibrated to large regulated institutions and says little about how the same roles are held by fewer people in a small one. An institution below that scale MUST adapt the structure and SHOULD record what it adapted, because an unadapted structure applied to a small institution produces roles nobody holds.
- *Missing.* The consolidation recorded at 6.2 has not yet been executed in the source book, which continues to assert the three former marks on its copyright page until a natural reprint. **This registry is the authoritative statement in the meantime**, and that is the reason 6.2 exists as a dated instrument rather than as a note.
- *Missing.* No charter, decision-rights assignment, cadence record or escalation record template is published with the deposited specification [11]. An institution can be told what a conformant charter must contain and

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cannot be handed one, so two institutions operating the framework will produce artifacts that are individually conformant and not comparable with each other. Recorded at this version because the deposit is what makes the gap visible.
- *Scope.* The framework specifies the governance function. It does not specify the competencies of the people in it, the training they require, or the individual credentials they hold.

4.7.15 Falsification condition. CADRE is falsified if institutions operating a chartered office with a documented single-accountable decision-rights assignment and a recorded cadence are found to escalate AI risks no earlier, and resolve them no more consistently, than institutions handling AI governance within existing risk committees without a distinct structure. The framework's claim is that a dedicated structure with named decision rights changes when and how a risk surfaces.

4.7.16 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
1.0	2026-08-30	Records the consolidation of the Governance Office Blueprint, the RACI-AI Matrix and the Governance Cadence Framework into this framework, per 6.2
2.0	2026-09-15	**Renamed** from the AI Governance Operating Model to CADRE, effective 1 September 2026. The identifier is unchanged and the former name is retained as the descriptor, per 6.9
2.0	2026-09-15	Mark status moves from `unmarked` to `trademark`. No clearance search has been completed and no registration has issued
2.0	2026-09-15	Status moves to `deposited` on the deposit of the CADRE specification [11], whose concept DOI the entry now carries
2.0	2026-09-15	Definition reconciled with that specification, which states the three parts as five testable artifacts
2.0	2026-09-15	Records a `missing` limitation for the four templates that specification does not publish
2.0	2026-09-15	The authority grant is extended to REG-09, resolving the question routed here at 8.2 of the Cross-Border AI Architecture Patterns specification [12]

4.8 REG-08 · Sharia AI Compliance Framework (SACF)

4.8.1 Identifier. REG-08.

4.8.2 Canonical name and mark status. The **Sharia AI Compliance Framework (SACF)™**. Trademarked. Short form after first mention: **SACF**. The spelling of the canonical name is fixed at Notation and MUST NOT be altered in citation.

4.8.3 Scope phrase. A dual-authority governance architecture for Islamic finance.

4.8.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22170143 [6]. The DOI given is the concept DOI and always resolves to the latest version. The source treatment is chapter 7 of the Enterprise Playbook [13], threaded through chapters 12, 13 and 14. This entry is updated in the registry version following that deposit.

4.8.5 Definition. SACF specifies the binding of a second, religiously authoritative approval power into AI lifecycle governance without splitting the institution into two governance systems.

Its central move is structural. The **Sharia Supervisory Board is treated as an authority source at the MESA Regulatory Floor**, alongside the civil supervisor, so that its determinations cascade down through the same machinery as civil regulation rather than running beside it in a parallel process. An institution operating SACF does not have a governance system and a Sharia process. It has one governance system with two authority sources at its floor.

Beneath that, the framework specifies a **Maqasid risk frame**, which classifies what an AI decision may harm in the terms the authority itself uses, and three threads into Operational Machinery.

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1. **Dual validation**, into step three of REG-02. A civil validator and a Sharia validator work as two independent tracks, and both MUST close before gate three approval under REG-06.
2. **The Halal data certification chain**, into REG-03. Source permissibility, transform integrity and certified use, attested end to end.
3. **Vendor screening**, into REG-04. The vendor's model conduct screened against the board's criteria as part of diligence.

Escalation is concurrent rather than sequential: an incident classified as Sharia-material under REG-05 routes to the Sharia Supervisory Board alongside civil escalation, not after it.

4.8.6 Purpose. To make an institution operating under two binding authorities able to satisfy both from one set of records, in a domain where the second authority's determinations are binding and the machinery for carrying them into an AI lifecycle does not exist.

4.8.7 Altitude. Regulatory Floor, as an authority source. SACF is the only framework in this registry that enters at the Regulatory Floor and then threads downward into three frameworks at Operational Machinery.

4.8.8 Inputs. The Sharia Supervisory Board's standing determinations and its approval authority as constituted by the institution. The Maqasid classification applied to the institution's AI decision types. The civil obligations already registered at the Regulatory Floor, since the framework's purpose is to compose the two rather than to substitute one.

4.8.9 Outputs. Sharia Supervisory Board approvals. Dual-validation records showing both tracks closed. Halal data certification chains. Vendor screening attestations. Sharia-materiality classifications on incidents.

4.8.10 Accountable role. The named Sharia liaison in the governance office, who is the institutional officer accountable for carrying the board's determinations into the lifecycle and for the completeness of the records above. **The Sharia Supervisory Board is an authority source and MUST NOT be recorded as the accountable role**, per 3.3. The board's authority is to determine; the liaison's accountability is to operate.

4.8.11 Evidence requirement. A record of the board's constitution and the scope of its binding authority. A dual-validation record per model where the branch applies, showing two independent closures and not one closure endorsed twice. Certification chains where the Halal chain is claimed. Screening attestations per screened vendor. Concurrent escalation records where a Sharia-material incident occurred.

4.8.12 Relationships. Sits at REG-01's Regulatory Floor as an authority source. Threads into REG-02 through dual validation, into REG-03 through the certification chain, and into REG-04 through vendor screening. Its closure is an entry criterion at gate three of REG-06. Escalates concurrently through REG-05. Full edges at 5.2.

4.8.13 Limitations.

- *Normative disclosure.* **No Sharia Supervisory Board has reviewed or endorsed this framework.** It is an engineering proposal for how binding Sharia authority can be integrated into AI lifecycle governance, offered for scholarly and institutional review. This clause is normative and MUST accompany any presentation of SACF until an endorsement exists and can be named.
- *Normative disclosure.* **The Halal data certification chain is author methodology.** No standard-setter recognizes such a chain as of the date of this registry. It MUST NOT be presented as an existing requirement of any Sharia standard.
- *Asserted.* That a Sharia Supervisory Board would accept the dual-validation construct, in which its validators operate as an independent track inside a lifecycle designed by engineers. No board has publicly accepted it.
- *Scope.* SACF specifies the architecture by which Sharia authority enters AI governance. **It does not make, and MUST NOT be read as making, any Sharia determination**, and it does not substitute for scholarly opinion on any question of permissibility.

4.8.14 Falsification condition. SACF is falsified if an institution operating the single-machinery constitution is found unable to satisfy both authorities from one set of records, so that a second and parallel record set has to be created for the Sharia authority in practice. The framework's central claim is that one governance system with two authority sources at its floor is sufficient for both. A parallel record set appearing under operation refutes that claim by observation rather than by opinion, which is what 3.6 requires, and it is observable by any party with access to the institution's records.

4.8.15 Change history.

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Entry version	Date	Change
1.0	2026-08-30	First registry entry, status `deposited`

4.9 REG-09 · Cross-Border AI Architecture Patterns

4.9.1 Identifier. REG-09.

4.9.2 Canonical name and mark status. Cross-Border AI Architecture Patterns™. Trademarked. Short form after first mention: **the Cross-Border Patterns**.

4.9.3 Scope phrase. Three deployment architectures for multi-jurisdiction AI, and the composition an estate reports when its workload classes resolve to more than one of them.

4.9.4 Specification status. `deposited`. Specification: https://doi.org/10.5281/zenodo.22285049 [12]. The DOI given is the concept DOI and always resolves to the latest version. Deposited on 15 September 2026, and **this entry is reconciled with that specification at this version, the specification governing under 11.3**. The source treatment remains chapter 9 of the Enterprise Playbook [13].

4.9.5 Definition. The Cross-Border Patterns specify **three** architectures for operating AI under the residency rules of more than one jurisdiction, selected in four steps.

Version 1.0 of this registry said four, and the deposited specification governs the count under 11.3, because how many patterns an architect may assign is a fact about the framework rather than about its place in the architecture [12]. Three patterns are selectable at the level of a workload class. **Hybrid is not a fourth pattern and MUST NOT be selected for a workload class**: it is the composition reported across an estate whose workload classes resolve to more than one of the three. The falsification condition at 4.9.14 already stated it this way at version 1.0, and this entry now agrees with it.

Step one enumerates the boundary set per jurisdiction: what data may not leave, what may move under licence, which authority examines what, and whether the exposure is administrative or criminal. **Step two** classifies each workload against that boundary set. **Step three** selects the pattern the boundaries permit.

Pattern	Applies when	Property
Sovereign Silo	No data may leave the jurisdiction	Full duplication per jurisdiction. Highest cost, highest control
Federated	Some movement is permitted under licence, raw data movement is not	Inference separated per jurisdiction, coordination central
Regional Hub	Transfer is permitted to a jurisdiction the others accept	One jurisdiction serves the region. Lowest cost, highest transfer exposure
Hybrid, *not selectable*	Reported when workload classes within one estate resolve differently	A composition of the three above, observed at estate level. **Not an assignment an architect makes**

Step four records the BOE Declaration: which boundary clauses were honoured, which optimization was chosen, and the routing evidence per request that proves the clauses held.

4.9.6 Purpose. To convert a residency constraint from a reason a deployment cannot proceed into a determinant of the architecture the deployment takes.

4.9.7 Boundary class, not altitude. REG-09 occupies no MESA altitude. It is one of two entries in that position and the only *subordinate* one, the other being REG-01, which defines the altitudes rather than standing outside them. It is the **worked instance of the Boundary Invariant applied to the data residency boundary class** [2]: the residency rule is the boundary, the topology is the optimization, and the four patterns are four optimizations under one uncrossable clause. This repositioning is recorded at 6.5 and is the reason the entry sits under the law rather than under an altitude.

4.9.8 Inputs. The residency rule set supplied by REG-03. The jurisdictions in scope and the transfer permissions in force between them. The workload classification. The examination scope of each authority involved.

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4.9.9 Outputs. The selected pattern with the boundary clauses it honours recorded against it. The deployment topology, which gate four of REG-06 approves. Residency attestations per jurisdiction. Routing evidence per request.

4.9.10 Accountable role. The named enterprise architect accountable for the deployment topology. Where the topology spans legal entities, the role is held in the entity that operates the platform, and the other entities are recorded as consulted.

4.9.11 Evidence requirement. A documented boundary set per jurisdiction, with each clause traceable to a primary source and dated. A workload classification against that set. A record of the selected pattern and the clauses that selected it. Routing evidence sufficient to show, per request, that no clause was crossed. **A pattern selected without a documented boundary set is a topology choice rather than a compliance position.**

4.9.12 Relationships. Worked instance of the Boundary Invariant, which is an engineering-family construct governed by [2]. **Operates under authority granted by REG-07, from version 2.0.** Version 1.0 recorded no authority grant to this framework, which its specification names at 8.2 as a real gap in the corpus rather than working around [12]; the grant closes it, and the requirement that specification places on an institution to record the grant its own governance supplies is unaffected. Consumes the residency rule set from REG-03. Supplies the topology to REG-06 at gate four. Its routing evidence is consumed by REG-05 during reconstruction. **No parent altitude.** Full edges at 5.2.

4.9.13 Limitations.

- *Asserted, and time-sensitive.* The jurisdiction-specific transfer rules cited in the source treatment age. **Every regulatory assertion MUST be re-verified against primary sources at the time of use**, and a boundary set carried forward from a prior deployment without re-verification is not a boundary set.
- *Asserted.* The worked cases in the source treatment are disclosed as composites. They illustrate the selection method and are not evidence that any named institution operates any of the patterns.
- *Scope.* The patterns govern where inference and data may sit. They do not specify the infrastructure, the network architecture or the contractual arrangements by which a topology is procured.

4.9.14 Falsification condition. The Cross-Border Patterns are falsified if a real boundary set requires a per-workload assignment in which some workload class can be served by none of Sovereign Silo, Federated or Regional Hub. Hybrid is not a fourth alternative for this purpose. It is defined at 4.9.5 as a composition of the other three, taken per workload class, and it therefore inherits their coverage rather than extending it. The claim under test is that the first three exhaust the topologies available under a residency constraint at the level of a single workload class, and a workload class none of them can serve would refute the enumeration directly. Stating the condition against Hybrid's composition rather than against the four as a set is deliberate, because a catch-all cannot be falsified.

4.9.15 Change history.

Entry version	Date	Change
1.0	2026-08-30	First registry entry
1.0	2026-08-30	Records the repositioning from peer framework to worked instance of the Boundary Invariant, per 6.5
2.0	2026-09-15	Status moves to `deposited` on the deposit of the Cross-Border AI Architecture Patterns specification [12], whose concept DOI the entry now carries
2.0	2026-09-15	Scope phrase and definition reconciled with that specification, which governs the count: **three** selectable patterns, with Hybrid as the composition of an estate
2.0	2026-09-15	The `missing` limitation recording no decision instrument is **closed** by Section 4 of that specification
2.0	2026-09-15	Receives an authority grant from REG-07, closing the gap that specification records at its 8.2

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4.10 Catalogue summary

ID	Canonical name	Mark	Altitude or class	Specification status	Accountable role
REG-01	MESA Framework	™	None. The frame itself	`deposited`	AI governance mandate executive
REG-02	MESA MRM Framework	™	Operational Machinery	`deposited`	Head of model risk
REG-03	AI Data Governance Framework	™	Technical Substrate	`deposited`	Data owner, per domain
REG-04	AI Vendor Risk Framework (AVRF)	™	Operational Machinery	`deposited`	Vendor relationship owner
REG-05	AI Incident Response Protocol (AIRP)	™	Operational Machinery	`deposited`	Incident lead
REG-06	Five-Gate Deployment Model	™	Operational Machinery, spine	`deposited`	One named person per gate
REG-07	CADRE	™	Strategic Compass	`deposited`	Head of the governance office
REG-08	Sharia AI Compliance Framework (SACF)	™	Regulatory Floor, authority source	`deposited`	Sharia liaison
REG-09	Cross-Border AI Architecture Patterns	™	Data residency boundary class	`deposited`	Enterprise architect

Nine marks, and every entry carries a deposited specification. Version 1.0 recorded eight marks and one deliberately unmarked framework, and five entries whose authoritative treatment was a book chapter. The count and the reasoning are recorded at 6.6 and the change at 6.9.

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5. The Relationship Register (Normative)

5.1 Relationship types

A relationship in this registry is a **typed, directed edge** from one framework to another. The vocabulary is closed. An entry MUST NOT assert a relationship of a type not defined here, and a new type MUST NOT be introduced except by a change to this section, because an open vocabulary is how two documents come to describe one dependency in two incompatible ways.

Type	Meaning	Asserted by
`instantiates`	The source framework occupies the named MESA altitude of the target	The subordinate framework
`grants-authority-to`	The source grants the authority under which the target operates	The granting framework
`supplies-evidence-to`	An output of the source is an input or a gate entry criterion of the target	The producing framework
`threads-into`	The source inserts a required branch into the target's own procedure, which the target then executes	The authority source
`absorbs`	The source has taken over the subject of a framework or instrument that no longer exists separately	The surviving framework
`supersedes`	The source replaces the target, which is retired	The surviving framework
`worked-instance-of`	The source is the application of a named law or boundary class to one domain	The instance

An edge MAY name more than one target where the same thing passes to each. Row 10 grants one authority to six frameworks and row 9 absorbs three retired names into one. Splitting either into single-target rows would assert

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one dependency several times, which 5.4 forbids. The count in this section is therefore a count of dependencies, not of endpoint pairs.

The direction of assertion is fixed by the third column and is not a stylistic choice. An edge is authored by the framework from which the thing moves: evidence moves from its producer, authority from its granter, a thread from the authority source that inserts it. Structural facts are authored by the subordinate or the survivor.

`consumes-evidence-of` is not an authored type. It is the derived converse of `supplies-evidence-to`, and any consumer view is generated by inverting the register rather than by writing a second set of rows. The reason is stated at 5.4.

5.2 The register

This table is normative. It is the authoritative statement of every dependency between the nine, and it is the source from which the relationship edges of `registry-v2.0.json` are generated. **Each dependency appears exactly once.**

#	From	Type	To	What passes
1	REG-02	`instantiates`	REG-01, Operational Machinery	Position in the maturity vector
2	REG-03	`instantiates`	REG-01, Technical Substrate	Position in the maturity vector
3	REG-04	`instantiates`	REG-01, Operational Machinery	Position in the maturity vector
4	REG-05	`instantiates`	REG-01, Operational Machinery	Position in the maturity vector
5	REG-06	`instantiates`	REG-01, Operational Machinery	Position in the maturity vector
6	REG-07	`instantiates`	REG-01, Strategic Compass	Position in the maturity vector
7	REG-08	`instantiates`	REG-01, Regulatory Floor	Registration as an authority source
8	REG-01	`absorbs`	Governance Maturity Model, retired	The maturity vector replaces the retired instrument
9	REG-07	`absorbs`	Governance Office Blueprint, RACI-AI Matrix, Governance Cadence Framework, all retired	Structure, decision rights and cadence become three sections of one framework
10	REG-07	`grants-authority-to`	REG-02, REG-03, REG-04, REG-05, REG-06, REG-09	Charter, decision rights, release authority
11	REG-02	`supplies-evidence-to`	REG-06	Validation and approval records, at G2 and G3, and the revalidation trigger register, at G5
12	REG-02	`supplies-evidence-to`	REG-05	Validation and approval records, for reconstruction
13	REG-03	`supplies-evidence-to`	REG-02	Data lineage and classification, at prevalidation
14	REG-03	`supplies-evidence-to`	REG-06	Classification, lineage and residency attestations, at G1
15	REG-03	`supplies-evidence-to`	REG-09	The residency rule set the patterns architect around
16	REG-03	`supplies-evidence-to`	REG-05	Classification and lineage records, for reconstruction
17	REG-04	`supplies-evidence-to`	REG-02	Vendor evidence where the model under validation is third-party
18	REG-04	`supplies-evidence-to`	REG-06	Vendor clearance at G1, contractual controls at G4
19	REG-04	`supplies-evidence-to`	REG-05	Vendor risk records and monitoring logs, for reconstruction
20	REG-05	`supplies-evidence-to`	REG-02	Revalidation triggers
21	REG-05	`supplies-evidence-to`	REG-06	Armed detection triggers at G5, and loop-back reopening G2

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#	From	Type	To	What passes
22	REG-06	`supplies-evidence-to`	REG-05	Gate passage records, for reconstruction
23	REG-07	`supplies-evidence-to`	REG-05	Escalation paths and escalation records
24	REG-08	`supplies-evidence-to`	REG-06	Dual-validation closure, at G3
25	REG-08	`supplies-evidence-to`	REG-05	Sharia-materiality classification and board approvals
26	REG-09	`supplies-evidence-to`	REG-06	The deployment topology, approved at G4
27	REG-09	`supplies-evidence-to`	REG-05	Routing evidence per request, for reconstruction
28	REG-08	`threads-into`	REG-02	The dual-validation branch at step three
29	REG-08	`threads-into`	REG-03	The Halal data certification chain
30	REG-08	`threads-into`	REG-04	Sharia vendor screening at diligence
31	REG-09	`worked-instance-of`	The Boundary Invariant, data residency class [2]	Residency is the boundary; topology is the optimization

Thirty-one edges. Every framework is the source of at least two, with the single exception recorded at 5.3. **The count is unchanged at version 2.0**: rows 10 and 11 each widened, one by a target and one by a subject, and no row was added or removed. Both changes dispose of a question a deposited specification routed here, and both are recorded in the change history of every entry they touch.

5.3 What the register shows

Three properties of the composed system are visible only in the register, and each is a claim the registry makes rather than a description it reports.

REG-05 is the only universal consumer. Seven of the seventeen `supplies-evidence-to` edges point at it, and they come from every other framework in the registry except REG-01, which supplies evidence to nothing. This is why an institution that operates the other eight without producing their evidence artifacts has a governance programme it cannot reconstruct from, and why the evidence requirement at 3.4 is a test rather than an aspiration. Incident reconstruction is the function that makes the evidence layer non-optional, because it is the only function that fails visibly and immediately when the evidence is absent.

REG-06 produces almost nothing of its own. Five-Gate is the target of six supply edges and the source of one. Its gate records attest that other frameworks' evidence was present and was relied on by a named person at a moment in time, and that is the whole of what it produces. That is why it is the spine: a spine carries load rather than generating it.

REG-01 is a target and almost never a source. MESA is the target of seven `instantiates` edges and the source of one `absorbs` edge, and it supplies evidence to nothing. This is correct and is the property that keeps the frame diagnostic. A frame that supplied inputs to the frameworks it measures would be measuring its own output, and REG-01 is therefore the single exception to the rule at 5.2 that every framework is the source of at least two edges.

REG-09 instantiates no altitude. It is the only subordinate entry that is the source of no `instantiates` edge, because it sits under the Boundary Invariant rather than under a MESA altitude. REG-01 is likewise the source of none, but it is the common target of all seven, which is a different position rather than a second exception. The absence is a property recorded at 4.9.7 and 6.5, not a gap in the register.

5.4 The closure rule: assert once, read from both ends

Every dependency MUST be asserted exactly once, in the direction fixed at 5.1, and MUST NOT appear a second time as its own converse.

This is the opposite of the arrangement a reader might expect, and the reason is the failure this registry was built to prevent. A dependency written down twice is a dependency that can be written down twice differently. The two statements will each be defensible, they will diverge under editing, and the divergence will be discovered by whoever relied on the wrong one rather than by an author.

So the converse is computed. A consumer view of REG-05, listing everything it reads, is generated by inverting the register. It is never authored, it cannot disagree with the register, and if the register changes the view changes with it.

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Validation therefore checks three things rather than symmetry, and these are the requirements the machine-readable layer enforces at 7.5.

1. Every edge names a source, and one or more targets that exist, either as registry identifiers or as declared external constructs.
2. **No dependency is authored twice.** An edge from A to B of an evidence, authority or thread type, where an edge of a converse sense also exists from B to A for the same subject, MUST fail validation.
3. Every edge's type is drawn from 5.1 and is asserted in the direction that section fixes.

Two edges may run between the same pair of frameworks in opposite directions where **different things move**, and the register carries two such pairs. Rows 12 and 20 run between REG-02 and REG-05: validation records travel one way for reconstruction, revalidation triggers the other. Rows 21 and 22 run between REG-05 and REG-06: gate records travel one way for reconstruction, armed triggers and the loop-back the other. A validation implementation MUST therefore compare the subject as well as the endpoints, and MUST NOT reject a legitimate two-way relationship as a duplicate.

5.5 Figure 2: composition at the gates

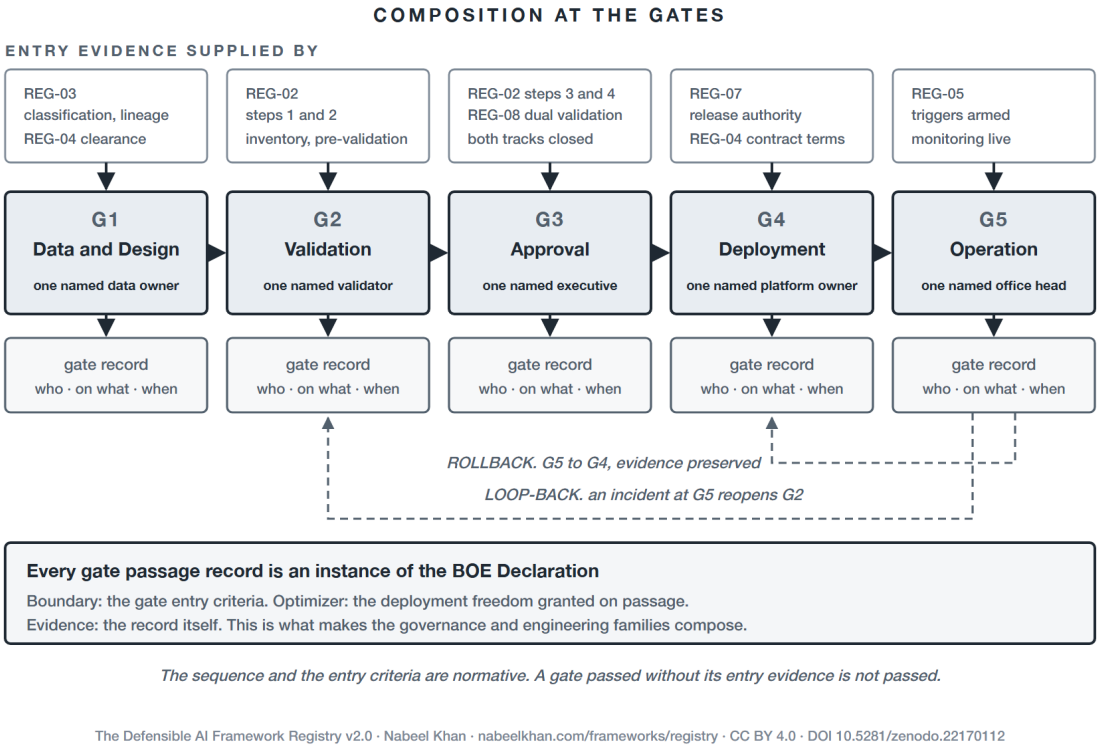


Figure 2. Composition at the gates. The Five-Gate Deployment Model consumes the outputs of data governance, vendor risk and model risk management as entry criteria, operates under authority granted by the Operating Model, and admits the dual-validation closure of SACF at gate three. Each gate names one accountable person and writes one gate passage record, which is an instance of the BOE Declaration. Rollback reverses gate five to gate four with evidence preserved; loop-back reopens gate two after an incident.

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6. Consolidation Notice (Normative)

This section is the dated, citable instrument of four decisions. It exists because a retirement, a rename or a supersession that is not recorded anywhere is not evidence of anything, and because the question of when a name stopped being used is one that only a dated record can answer.

Effective date of every decision recorded here: 30 August 2026, the publication date of registry version 1.0.

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6.1 The state being consolidated

Twelve trademarks were asserted on the copyright page of the Enterprise Playbook, First Edition, 2026 [13]. They are listed at Appendix A.1 in the form in which they were asserted. Four decisions are recorded below. Their combined effect is stated at 6.6.

6.2 Consolidation: three frameworks become one

The Governance Office Blueprint, the RACI-AI Matrix and the Governance Cadence Framework are consolidated into the AI Governance Operating Model (REG-07), in which they become the Structure, Decision rights and Cadence sections respectively.

The three answered three questions about one subject: how the governance function is organized, who decides within it, and how often it meets. Those are three sections of one framework. Asserting them as three separate marks inflated the count of a portfolio while dividing a subject that an institution encounters as one. A decision-rights matrix has no operational meaning without a structure to assign the rights within, and a cadence has none without both.

The three former names remain valid for citation of the work published under them. They MUST NOT be used for new work, and any surface presenting them as current frameworks is out of date with this registry.

6.3 Supersession: the Governance Maturity Model

The Governance Maturity Model published in the Enterprise Playbook [13] is superseded by the evidence-based maturity vector specified in the MESA Framework specification [1].

The two instruments answered the same question and MUST NOT be operated in parallel. The vector supersedes because it reports per altitude rather than per organization, and because each of its levels carries an evidence requirement that can be tested rather than asserted.

Institutions holding a prior assessment under the retired model MAY map their result to the vector using the mapping table published in the MESA specification. **That mapping is informative, not normative.** A mapped result indicates a starting position and does not constitute an assessment, because the prior model did not require per-altitude evidence and in particular performed no independent assessment of the Technical Substrate. Under MESA's enforcement constraint, a mapped result is therefore an upper bound on the institution's documented posture and says nothing about its enforced one. That asymmetry is the reason for the supersession rather than an artifact of it.

6.4 Rename: AI Data Governance Stack becomes AI Data Governance Framework

The framework asserted as the AI Data Governance Stack is renamed the AI Data Governance Framework™ (REG-03).

The former name implied a technology stack. The content is a control taxonomy, and the new name aligns the family, in which the peer frameworks are already named Framework rather than Stack.

Two consequences are recorded because they are consequences rather than intentions. The former name has prior use from the First Edition and the new name has none as of the date of this registry, and the question of whether asserting the new name affects rights accrued under the old one is a question for a trademark agent that this registry does not answer. The former name is recorded at Appendix A.2 and remains valid for citation of the earlier work.

6.5 Repositioning: Cross-Border AI Architecture Patterns

Cross-Border AI Architecture Patterns™ (REG-09) is repositioned from a peer framework of MESA to the worked instance of the Boundary Invariant applied to the data residency boundary class.

The name is unchanged. What changes is where the framework sits in the architecture, and therefore what it inherits. Residency is the boundary, the topology is the optimization, and the four patterns are four optimizations under one uncrossable clause. The repositioning is what allows a residency decision to carry a BOE Declaration in the same shape as a gate record, which is what makes routing evidence usable during an incident reconstruction rather than merely available.

The consequence is that REG-09 occupies no MESA altitude. This is recorded as a property of the entry at 4.9.7. It is the only subordinate framework in that position. REG-01 is also outside the altitudes, but because it defines them rather than because it escapes them.

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6.6 The count

Before, as asserted [13]	After, as registered here	Change
MESA Framework™	REG-01, MESA Framework™	Kept. Absorbs the maturity model
MESA MRM Framework™	REG-02, MESA MRM Framework™	Kept
AI Data Governance Stack	REG-03, AI Data Governance Framework™	Renamed, and repositioned as the data boundary class
AI Vendor Risk Framework (AVRF)™	REG-04, AI Vendor Risk Framework (AVRF)™	Kept
AI Incident Response Protocol (AIRP)™	REG-05, AI Incident Response Protocol (AIRP)™	Kept. Reframed around reconstruction
Governance Office Blueprint	Consolidated into REG-07	Retired as a separate name
RACI-AI Matrix	Consolidated into REG-07	Retired as a separate name
Governance Cadence Framework	Consolidated into REG-07	Retired as a separate name
Five-Gate Deployment Model™	REG-06, Five-Gate Deployment Model™	Kept. Promoted to the lifecycle spine
Governance Maturity Model	Superseded by the MESA maturity vector	Retired
Sharia AI Compliance Framework (SACF)™	REG-08, Sharia AI Compliance Framework (SACF)™	Kept
Cross-Border AI Architecture Patterns™	REG-09, Cross-Border AI Architecture Patterns™	Kept. Repositioned under the Boundary Invariant
Did not exist	The Boundary Invariant	Engineering family. Deliberately unmarked [2]
Did not exist	The BOE Declaration	Engineering family. Deliberately unmarked [2]

Twelve asserted marks, less three consolidated, less one superseded, plus one consolidated framework that is deliberately unmarked, gives **eight marks and one unmarked framework: nine registry entries.**

6.7 What this notice does not do

It does not amend the source book. The Enterprise Playbook First Edition continues to assert twelve marks on its copyright page and is not reprinted for this. The corrections are queued for a natural reprint. **This registry is the authoritative statement in the meantime**, which is the reason it exists as a dated instrument rather than as an erratum.

It does not resolve any question of trademark registration. Whether any of the eight remaining marks is registered, registrable or enforceable in any jurisdiction is outside this registry, which records usage rather than rights. **No mark in this registry carries the registered-mark symbol, and none MUST until a registration certificate issues for that mark in that jurisdiction.**

It does not retire the retired names from the record. A retired name remains the correct citation for the work published under it. Appendix A carries every former name for that purpose.

6.8 Figure 3: the consolidation

6.9 What version 2.0 records (Normative)

6.6 is the dated instrument of what happened on 30 August 2026 and is not rewritten here, because a consolidation notice that is edited afterwards is not evidence of anything. This subsection records what version 2.0 changes, on the same terms.

One rename. *AI Governance Operating Model* becomes **CADRE**, effective 1 September 2026. The identifier REG-07 is unchanged, as 11.2 requires, and **the former name is retained as the framework's descriptor rather**

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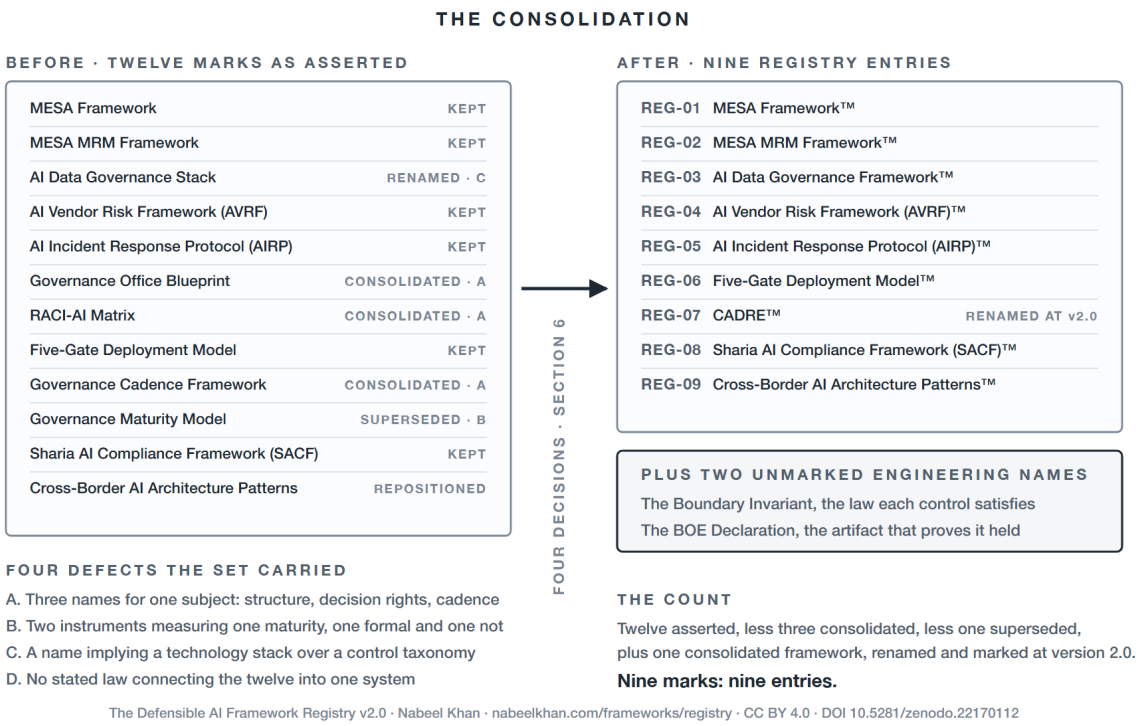


Figure 3. The consolidation. On the left, twelve marks as asserted, with the three defects marked: three names for one subject, two instruments for one measurement, and a name implying a technology stack. On the right, nine registry entries under one architecture, with the two deliberately unmarked engineering-family names that supply the law the nine operate under.

than withdrawn from use. That distinguishes it from the OTHER rename at A.2 and from the retired names at A.3: those are names no longer used, and this is a name that continues in a different role. The canonical first mention is *CADRE™, the AI Governance Operating Model*. **This rename is why version 2.0 is a major version**: 11.1 makes a canonical name change a major event, because a claim written against the former name may no longer be conformant against the new one. Work published under the former name remains citable and is not an error, per the rule at the head of Appendix A. The derivation of the name is at [11].

One mark added. REG-07 moves from `unmarked` to `trademark`, taking the governance family to nine marks and none deliberately unmarked. The Boundary Invariant, the BOE Declaration, PEVG and PARA are engineering-family constructs and remain unmarked for the reasons at 2.3. **No clearance search has been completed for CADRE**, which 4.7.4 and 10.6 record as open.

Five specification statuses closed. REG-02, REG-03, REG-05, REG-07 and REG-09 move from `source-treatment` to `deposited`, each carrying the concept DOI of its specification [8]-[12]. **Every entry in this registry now points at a standalone specification**, which was the state 11.4 recorded as expected. Each of the five entries is reconciled with its specification, the specification governing under 11.2 and 11.3, and each reconciliation is named in that entry's change history rather than applied silently.

Two questions disposed of. Both were routed here by a deposited specification rather than settled by it, which is what 5.4 requires of a document that would otherwise author an edge about itself.

1. **The gate five supply edge.** [8] 6.3 records that Five-Gate names the revalidation trigger register as a gate five entry criterion and attributes it to MESA MRM [3, 3.6], while row 11 carried the supply at gates two and three only. **The register was incomplete**, and row 11 now carries gate five. The alternative, authoring a second REG-02 to REG-06 row, is forbidden by 5.4.
2. **The authority grant to REG-09.** [12] 8.2 records that the register grants authority to five frameworks and none to REG-09, and declines to work around it. 2.4 holds that evidence cannot be produced without authority; REG-09 produces evidence at rows 26 and 27. **The gap was in this register**, and row 10 now carries REG-09 as a sixth target. A reader will find [11] stating that CADRE grants authority to five others: that was true of the register published at version 1.0, and is superseded here under 11.3.

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One correction to version 1.0. Item 1 of 11.4 stated that REG-01 and REG-07 might move from `source-treatment` to `deposited`. REG-01 was already `deposited` at version 1.0 and could not move from a status it did not hold. The artifact that item meant is the MESA assessment instrument, which is not a registry entry and does not change REG-01's status. Corrected at 11.4 and recorded here rather than fixed silently.

What this subsection does not do. It does not amend 6.6, it does not retire any former name from the record, and it resolves no question of trademark registration. **No mark in this registry carries the registered-mark symbol, and none MUST until a registration certificate issues for that mark in that jurisdiction.**

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7. The Machine-Readable Layer (Normative)

7.1 Why the layer exists

A registry that is readable only by people will be restated by a machine interface, and the restatement will drift. The drift will not announce itself: the document and the interface will each remain internally coherent while giving different answers to the same question, and the discrepancy will be found by whoever relied on the wrong one.

This registry therefore publishes its entries once, in two representations generated from the same content, and specifies the precedence between them.

7.2 The files

File	What it is
`registry-v2.0.json`	One object per framework, carrying the fields of 3.1 in machine-readable form, plus the relationship edges of 5.2
`registry-schema-v2.0.json`	The JSON Schema the above MUST validate against. Published so that a third party can validate the registry rather than trust it. **Every file valid against the 1.0 schema is valid against this one**: the single change is a relaxation, permitting a marked entry to record why its mark was taken where 1.0 required that field to be empty. The version number follows the registry because these two files are a pair

Both files are part of the deposit and are licensed CC BY 4.0 with the rest of it.

7.3 Precedence

This document governs. Where `registry-v2.0.json` and Section 4 disagree, Section 4 is correct and the JSON is defective. The JSON is a representation of the registry and is not a second registry.

Stating the precedence in this direction is deliberate. The opposite arrangement, in which the data file governs and the prose describes it, is workable for a data product and wrong for a specification, because the fields that carry the registry's actual content are the definitions, the limitations and the falsification conditions, and those are paragraphs.

7.4 Requirements on any interface serving this registry

An interface that serves this registry, including a machine interface answering questions about the frameworks, MUST meet all of the following.

1. **It MUST serve `registry-v2.0.json` unmodified**, or serve content generated from it, and MUST NOT serve a separately authored statement of the same facts.
2. **It MUST cite the registry version and the DOI** in any response asserting a name, definition, relationship or status from it. A response that asserts a framework's definition without stating which version of the registry it came from cannot be checked, and an unverifiable answer about provenance is the failure mode this registry exists to avoid in its own domain.

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3. **It MUST NOT present a `source-treatment` or `instrument-pending` framework as having a deposited specification.** The status field exists to be reported, not to be smoothed over.
4. **It MUST reproduce the two normative honesty clauses of REG-08 at 4.8.13** in any response describing SACF. Those clauses are part of the framework's statement, not commentary on it.
5. **It SHOULD expose the schema alongside the data**, so that a consumer can validate what it received.

7.5 Validation requirements

`registry-v2.0.json` MUST satisfy all of the following, and a build that produces a file failing any of them MUST NOT be published.

1. It validates against `registry-schema-v2.0.json`.
2. It contains exactly nine framework objects, with identifiers REG-01 through REG-09, each unique.
3. Every `specification_status` value is one of the three defined at 3.2, and every object with status `deposited` carries a resolvable DOI while every object with another status carries none. **At version 2.0 all nine carry `deposited`.** The other two values remain in the vocabulary, because a future entry may be added under either.
4. Every relationship edge names one or more targets that exist, either as registry identifiers or as declared external constructs.
5. **No dependency is authored twice**, per 5.4. Where an edge of a converse sense exists between the same pair of frameworks, the two MUST carry different subjects, and a validation implementation MUST compare the subject and not only the endpoints.
6. The edge count equals the number of rows in the register at 5.2, and every edge type is asserted in the direction fixed at 5.1.
7. Every trademarked name carries `"mark": "trademark"` and every unmarked name carries `"mark": "unmarked"` with a stated rationale. **At version 2.0 every object carries `trademark`**, so the unmarked branch of this rule binds no entry in this version and is retained because a future entry may be added unmarked. A marked entry MAY carry a rationale and REG-07 does, for the reason at 4.7.4. **No object carries a registered-mark symbol.**

7.6 Figure 4: the machine-readable layer

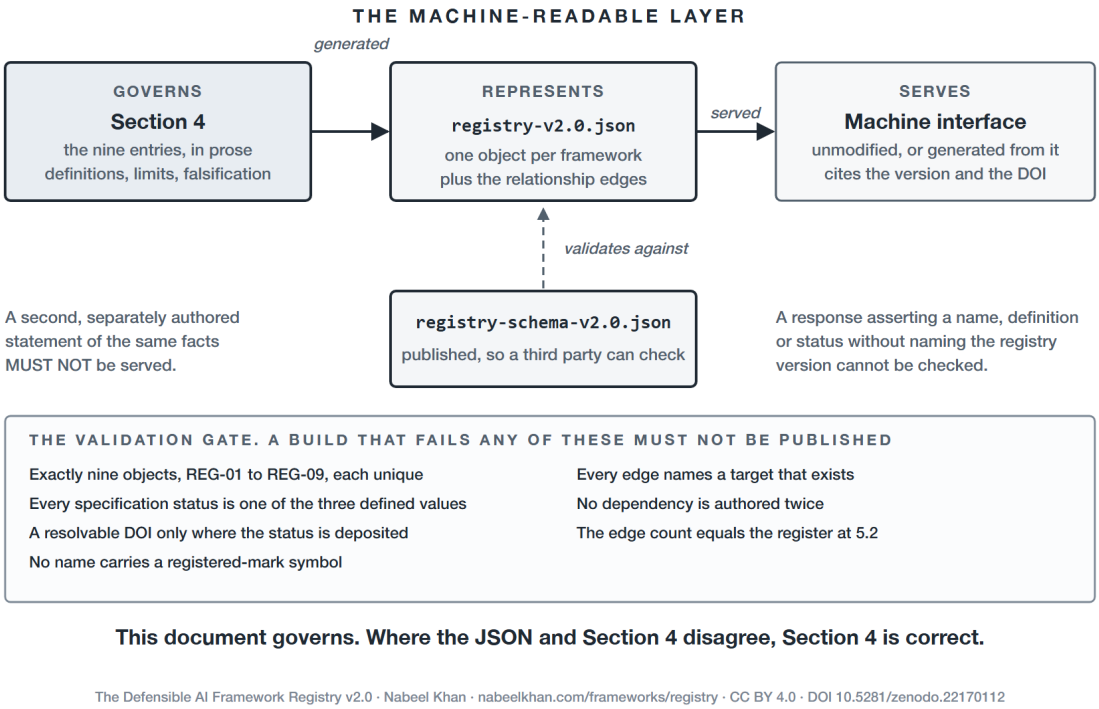


Figure 4. The machine-readable layer. Section 4 of this document is the governing statement. The JSON is generated from it, validated against a published schema, and served by any machine interface without a second authored statement of the same facts. The validation gate enforces closure of every relationship edge and the correspondence between the document and the data, so that drift fails a test rather than reaching a reader.

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8. Conformance (Normative)

8.1 Two claims, both self-declared

This registry supports exactly two conformance claims. Both are self-declared. **No conformity assessment body operates against this registry, and no third party is accredited to assess a claim made under it.**

8.2 Citation conformance

A document, page, deck or machine response is **citation conformant** with this registry if all of the following hold.

1. Every framework it names uses the canonical name from Section 4 at first mention, with the mark and the scope phrase where the entry carries them, and the short form thereafter.
2. It cites the registry version and DOI where it asserts a definition, relationship or status.
3. It does not present a retired name from Appendix A as a current framework.
4. It does not apply the registered-mark symbol to any mark in this registry.
5. Where it describes SACF, it reproduces the two honesty clauses of 4.8.13.

Citation conformance is a property of a document and is cheap to check, which is the point of stating it.

8.3 Alignment conformance

An institution is **alignment conformant** with this registry, for a stated set of entries, if for every entry in that set all of the following hold.

1. A single named person holds the accountable role, per 3.3.
2. Every evidence artifact named in the entry's evidence requirement is being produced, per 3.4.
3. Every relationship in which the entry is a consumer is satisfied by an actual input, rather than by an intention to obtain one.

An institution MUST state the set of entries it claims, and MUST NOT claim conformance for the registry as a whole unless it claims all nine. Partial claims are expected and are the normal case, since REG-08 applies only to institutions under Sharia authority and REG-09 only to institutions operating across jurisdictions.

A published alignment claim SHOULD carry: the entries claimed, the entries deliberately not claimed with the reason, the date of the evidence relied on, and the registry version claimed against.

8.4 What a conformance claim does not mean

A claim under 8.2 or 8.3 does not mean any of the following, and any presentation implying otherwise is a misuse of this registry.

- **It does not mean the institution is compliant** with any statute, supervisory expectation or standard.
- **It does not mean the frameworks are effective.** Section 10 states what has not been tested.
- **It does not mean the institution has been assessed.** Nobody assessed it. It declared.
- **It does not mean a certification exists.** None does.

8.5 Withdrawal of a claim

An institution whose evidence artifacts stop being produced is no longer alignment conformant for the affected entry from the date production stopped, not from the date the lapse is discovered. **A conformance claim carries the date of the evidence, and evidence has a currency**, which means a claim that is not refreshed becomes a historical statement rather than a current one.

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9. Related Work

This section positions the registry rather than the frameworks it lists. Each entry in Section 4 positions its own framework.

9.1 Management system standards

ISO/IEC 42001:2023 [17] specifies a management system for artificial intelligence, and the NIST **AI Risk Management Framework** [18], with its generative AI profile [19], specifies a risk management structure with functions and categories. Both operate at institutional altitude. Neither is a registry: both specify what an institution should do, and neither fixes the names, versions and relationships of a body of separately authored frameworks, which is this document's function. An institution certified to ISO/IEC 42001 could operate every framework in this registry, none of them, or some of them, and this registry does not change that.

The MESA Framework specification carries clause-level mappings to both [1]. Those mappings are the author's readings, and neither issuing body has reviewed or endorsed them.

9.2 Supervisory guidance on model risk

Two supervisory instruments bear directly on REG-02. In Canada, **OSFI Guideline E-23, Model Risk Management (2027)** [20], published 11 September 2025 and effective 1 May 2027, extends model risk expectations to artificial intelligence and machine learning models and to models and data sourced from third parties. In the United States, **SR 26-2, Revised Guidance on Model Risk Management** [21], issued 17 April 2026, supersedes and replaces SR 11-7, the 2011 guidance that had been the reference for fifteen years.

This registry states neither guideline's requirements and does not claim that REG-02 satisfies either. The relationship is one direction only: an institution subject to either instrument will find that REG-02 organizes the discipline the instrument expects, and MUST perform its own crosswalk against the primary source in force. The absence of that written crosswalk is recorded as a limitation at 4.2.13 rather than left to be discovered.

9.3 Incident response

NIST SP 800-61 Revision 3 [22], published April 2025, recommends incident response practice as a Cybersecurity Framework community profile. It is the mature instrument in its field and REG-05 does not compete with it. The difference is in what the two are built to reconstruct. A cyber incident response process establishes what an attacker did. REG-05 establishes what a model decided, on what data, under which policy version, and which person had approved it, which is a reconstruction that requires evidence written at decision time by frameworks other than the incident protocol itself. An institution operating both will find REG-05 consuming evidence that no cyber process produces.

The absence of an alignment between REG-05's classifications and the public incident repositories, the **OECD AI Incidents and Hazards Monitor** [26] and the **AI Incident Database** [27], is recorded as a limitation at 4.5.13. Without it, an institution's incident record cannot be compared to any population outside itself.

9.4 Sharia governance

The standing base instrument is **IFSB-10, Guiding Principles on Shariah Governance Systems for Institutions Offering Islamic Financial Services** [23], issued December 2009, supplemented by **IFSB-31, Guiding Principles for Effective Supervision of Shariah Governance** [24], issued July 2025. **Neither addresses artificial intelligence.** IFSB-31 was examined for this purpose and contains no treatment of AI or machine learning.

AAOIFI has issued no standard or exposure draft on artificial intelligence as of the date of this registry. Its recorded activity in the area is a collaboration memorandum for generative AI training programmes, signed 27 February 2025 [25], and conference programming on the theme.

The field is therefore open in the specific sense that matters to REG-08: no standard-setter has specified how binding Sharia authority integrates into an AI system lifecycle. That is what makes REG-08 an engineering proposal offered for scholarly and institutional review rather than an implementation of an existing standard, and it is why the honesty clauses at 4.8.13 are normative rather than editorial.

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9.5 Framework registries in other fields

Registries of named constructs are unremarkable in mature fields and were the model for this one. A registry succeeds when the identifier survives every rename and reorganization applied to the thing it identifies, which is why 3.1 fixes the identifier first and the name second. The convention is borrowed rather than invented, and the registry claims no novelty in the form.

9.6 The engineering-family specification

A Pattern Language for Production LLM Platforms [2] is the closest related work and is not a competitor to this registry. It is the governing document of the other family, it supplies the Boundary Invariant and the BOE Declaration that Section 2 rests on, and the precedence rule at 2.6 exists so that the two documents cannot both claim the same relationship. Where an institution operates both families, the registry describes the governance obligations and the pattern language describes the platform that must enforce them.

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10. Limitations and What Has Not Been Tested

10.1 The registry describes a composition that has not been observed

No institution unconnected to the author has been observed operating the composed set. Each framework has been applied; the nine as an architecture have not been evaluated by anyone outside the author's practice, and no third party has published a result obtained under them.

This is the single most important limitation in the document and it is stated first. Everything in Section 2 is an architectural claim. The registry is coherent, which is a property that can be checked from the document, and it is not thereby correct, which is a property that cannot.

10.2 The entries are not specifications

Five of the nine frameworks have no standalone specification, their authoritative treatment being a book chapter. For those five the entry is the most complete normative statement available in citable form, and for the five it is a summary of a chapter rather than a specification of a framework. **A reader implementing from an entry alone is implementing from a summary**, and the entry says so through its `specification_status` field rather than leaving the reader to infer it.

10.3 The limitations of the frameworks are inherited, not resolved

Each entry records what its framework asserts without evidence and what it is missing. This registry resolves none of them. Composing nine frameworks does not evidence any of them, and a reader should not read the coherence of the architecture as support for any member of it.

10.4 The adoption sequence is untested

The architecture states dependencies. It does not state that adopting the frameworks in dependency order is optimal, or that an institution adopting REG-06 before REG-02 will fail. **The dependency graph is a statement about what a framework needs in order to operate, not a statement about what a programme should sequence.** No adoption sequence has been evaluated.

10.5 The evidence requirements have not been costed

Every entry requires artifacts. The registry does not state what producing them costs, at what institutional scale the cost becomes prohibitive, or which requirements are the expensive ones. An institution reading the evidence requirements as a checklist may find that satisfying all of them is not proportionate to the risk of its AI estate, and the registry offers no proportionality guidance.

10.6 Three frameworks depend on an authority that has not acted

REG-08 requires the acceptance of Sharia Supervisory Boards, and none has publicly given it. REG-06 uses a name whose availability has not been cleared, and **REG-07 now does the same**: CADRE is claimed as a mark

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from this version and **no trademark clearance search has been completed for it**, which 4.7.4 records as open rather than settled. All three are recorded in their entries. Neither is resolved here, and neither can be resolved by the author acting alone, which is the property that distinguishes them from the other limitations in this section.

10.7 The registry is a single-author instrument

Every framework in it has one author, and so does the registry that composes them. A registry maintained by its own beneficiary carries an obvious structural conflict: the instrument that decides which frameworks belong to the discipline is written by the person whose frameworks they are.

The mitigations are stated rather than assumed. The identifiers cannot be renumbered, so nothing can be quietly removed. The consolidation notice is dated, so a retirement cannot be backdated. The falsification conditions are published, so the frameworks can be attacked on stated terms. The machine-readable layer is validated against a published schema, so a third party can check the registry rather than trust it. **None of these makes the instrument independent.** They make it auditable, which is a different and lesser property, and the registry claims only the lesser one.

10.8 What would falsify the registry

The registry is falsified separately from any framework in it. Each entry states its own falsification condition at its fourteenth field, and those conditions are not restated here. The registry itself, as an architecture, is falsified by any of the following.

1. **A framework in the set that cannot be placed.** If a framework required by regulated institutions to govern AI is found to occupy no MESA altitude and to be no instance of the Boundary Invariant, then the architecture does not span its own domain and Section 2 is incomplete rather than merely unproven.
2. **A dependency that runs the wrong way.** If institutions are found to require the outputs of REG-06 in order to operate REG-02, rather than the reverse, the spine is inverted and the register at 5.2 misdescribes the system it claims to describe.
3. **A composition that does not compose.** If gate records, validation records, residency attestations and screening attestations are found in practice not to be joinable into a reconstruction, because the BOE Declaration does not in fact produce evidence in one shape, then the claim at 2.3 fails and the nine are adjacent rather than composed.
4. **Consolidation that loses content.** If institutions operating REG-07 are found unable to express a structure, decision right or cadence that the three retired frameworks could express, the consolidation at 6.2 destroyed content rather than removing duplication.
5. **A maturity profile that adds nothing.** If per-altitude profiles are found to give institutions no better information than a single grade, REG-01 fails and, since Section 2 rests on it, so does the frame in which the other eight are placed.

The first three would refute the architecture. The fourth would refute one decision recorded in Section 6. The fifth would refute a proposition this registry borrows rather than owns, and the registry would fall with it.

10.9 Research questions

The registry is written to make the following answerable. It answers none of them.

1. Do institutions reporting a per-altitude maturity profile take different remediation decisions than institutions reporting a single grade, and are the decisions better?
2. Is the dependency graph at 5.2 the graph institutions actually encounter, or does adoption in practice reveal edges the register does not carry?
3. Does a gate with one named accountable person produce a different deployment outcome than an equivalent automated check with none?
4. What does producing the full evidence set cost, at what institutional scale does it stop being proportionate, and which of the requirements carries most of the cost?
5. Would a Sharia Supervisory Board accept constitution as an authority source inside a civil governance machinery, and if not, what constitution would it accept?
6. Does an incident reconstruction built from decision-time BOE records satisfy an external authority in a way that conventional logs do not?

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A reader able to answer any of these from real deployment evidence is invited to make contact through the ORCID record in the front matter. The registry has no dataset, and question four in particular is answerable only from institutions that have done the work.

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11. Versioning and Change Control

11.1 What each version number means

The major number changes when a change to this registry could make a previously conformant claim non-conformant: a canonical name changes, an entry is retired, a relationship type is removed or redefined, or a conformance requirement in Section 8 is added or tightened.

The minor number changes for additions and clarifications that cannot: a new entry, a new relationship edge, a corrected limitation, a specification status moving from `source-treatment` or `instrument-pending` to `deposited` when a DOI is minted.

11.2 What MUST accompany a change

- **A new entry** MUST carry all fifteen fields of 3.1 and its edges MUST close per 5.4.
- **A rename** MUST record the former name at Appendix A and MUST NOT change the identifier.
- **A retirement** MUST record the identifier as retired. **The identifier MUST NOT be reused.**
- **A status change to `deposited`** MUST carry the concept DOI of the deposited specification, and the entry's definition MUST then be reconciled with that specification, with the specification governing.
- **Every change** MUST appear in `CHANGELOG.md` classified as normative or editorial.

11.3 The registry and the specifications it points at

Where a framework's specification is deposited, **the specification governs the framework and this registry governs the framework's place in the architecture.** A disagreement about what MESA requires is resolved by the MESA specification. A disagreement about which altitude REG-04 instantiates is resolved here. Neither document may resolve the other's question, and a version of either that begins to do so is a defect.

11.4 Expected changes to version 2.0

Recorded so that a reader can distinguish a planned change from a correction.

What version 1.0 expected, and what became of it. Item 1 of this section at version 1.0 read that REG-01 and REG-07 might move from `source-treatment` to `deposited`. **It was wrong about REG-01**, which was already `deposited` at version 1.0 and could not move from a status it did not hold; the artifact meant was the MESA assessment instrument, which is not a registry entry. REG-07 did move, along with four others, and **every entry now carries a deposited specification**. The correction is recorded at 6.9 rather than made silently.

1. **Templates and instruments are published against entries that record them as missing.** The charter, decision-rights, cadence and escalation templates of REG-07, the gate record templates of REG-06 and the MESA assessment instrument are each named in an entry as a gap. Publishing any of them is a minor version of this registry and closes a limitation, not an entry.
2. **The engineering-family constructs PEVG and PARA are deposited separately.** PEVG is at [4] and PARA at [5]. They are not entries in this registry, which covers the governance family only; they are cited here so a reader can reach them.
3. **Entries gain evidence.** Each entry's limitations field records what is asserted and what is missing. Every one of those is a candidate for removal by evidence rather than by editing.
4. **A clearance search is run against CADRE, Five-Gate and the other marks, or is recorded as still not run.** 10.6 states the position today. A search that issued a result either way would change three entries and this section.

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12. References

Every reference below was verified against a live primary source on 29 August 2026, using a deliberate failing control for each check, except where a note records otherwise. A control that cannot fail proves nothing, so each check was accompanied by an invented identifier or URL of the same shape that was confirmed to fail.

[1] Khan, N. A. (2026). *The MESA Framework: A Four-Altitude Diagnostic Model for Institutional AI Governance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109836

[2] Khan, N. A. (2026). *A Pattern Language for Production LLM Platforms: Governed Routing, Agent Orchestration, and AI-Native Delivery. Pattern Language Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109864

[3] Khan, N. A. (2026). *The Five-Gate Deployment Model: A Deployment Discipline for AI Systems. Deployment Model Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170122

[4] Khan, N. A. (2026). *PEVG: Planner, Executor, Verifier, Generator. Four Contracts, and the Two Boundaries a Verifier Does Not Both Hold. Pattern Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170132

[5] Khan, N. A. (2026). *PARA: Perception, Action, Reasoning, Adaptation. Four Faculties, Four Authority Types, and the Registry Entry That Turns a Faculty into a Contract. Pattern Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170139

[6] Khan, N. A. (2026). *The Sharia AI Compliance Framework: A Dual-Authority Governance Architecture for Islamic Finance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170143

[7] Khan, N. A. (2026). *The AI Vendor Due-Diligence Questionnaire: AVRF Instrument for Third-Party AI Systems, Models and APIs. Questionnaire Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170146

[8] Khan, N. A. (2026). *The MESA MRM Framework: A Six-Step Model Risk Management Discipline for AI Systems. Framework Specification, Version 1.0.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285045

[9] Khan, N. A. (2026). *The AI Data Governance Framework: A Five-Stage Control System for the Data Boundary in AI Systems. Framework Specification, Version 1.0.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285047

[10] Khan, N. A. (2026). *The AI Incident Response Protocol: Six Stages, and the Evidence a Reconstruction Requires. Protocol Specification, Version 1.0.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285057

[11] Khan, N. A. (2026). *CADRE: Charter, Authority, Decision Rights, Records, Escalation. The Governance Function Specified as Five Artifacts. Framework Specification, Version 1.0.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285052

[12] Khan, N. A. (2026). *Cross-Border AI Architecture Patterns: Three Patterns, and Why the Fourth Is Not a Choice. Pattern Specification, Version 1.0.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22285049

[13] Khan, N. A. (2026). *AI Governance and Compliance Frameworks for the Middle East: The Enterprise Playbook.* First Edition. iSystematic Inc. ISBN 978-1-0678960-1-0

[14] Khan, N. A. (2026). *LLM Systems in Production: Cloud-Native Patterns for AI Engineers.* iSystematic Inc. ISBN 978-1-0678317-1-4

[15] Khan, N. A. (2026). *Prompt Systems and Agent Orchestration: Engineering Multi-Model AI Workflows.* iSystematic Inc. ISBN 978-1-0678317-2-1

[16] Khan, N. A. (2026). *DevOps for AI-Native Platforms: Building, Governing, and Scaling AI Infrastructure.* iSystematic Inc. ISBN 978-1-0678317-3-8

[17] International Organization for Standardization and International Electrotechnical Commission (2023). *ISO/IEC 42001:2023, Information technology. Artificial intelligence. Management system.* First edition, December 2023.

[18] National Institute of Standards and Technology (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* NIST AI 100-1, 26 January 2023. Current as of the date of this registry; a revision is in progress and not published.

[19] National Institute of Standards and Technology (2024). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile.* NIST AI 600-1, 26 July 2024. A companion to [18], not a replacement.

[20] Office of the Superintendent of Financial Institutions, Canada (2025). *Guideline E-23, Model Risk Management (2027).* Published 11 September 2025, effective 1 May 2027. Scope includes artificial intelligence and machine learning models and models or data sourced from third parties.

[21] Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency and Federal Deposit Insurance Corporation (2026). *SR 26-2, Revised Guidance on Model Risk Management.* Issued 17 April 2026. Supersedes and replaces SR 11-7 (2011) and SR 21-8 (2021).

[22] National Institute of Standards and Technology (2025). *Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile.* NIST SP 800-61 Revision 3, April 2025. Supersedes Revision 2.

[23] Islamic Financial Services Board (2009). *IFSB-10, Guiding Principles on Shariah Governance Systems for Institutions Offering Islamic Financial Services.* December 2009.

[24] Islamic Financial Services Board (2025). *IFSB-31, Guiding Principles for Effective Supervision of Shariah Governance.* July 2025. Examined for this registry; contains no treatment of artificial intelligence or machine learning.

[25] Accounting and Auditing Organization for Islamic Financial Institutions (2025). *AAOIFI signs collaboration MoU with the UK-based ICMA Digital Academy to introduce Gen-AI training programs in IFIs.* Announcement, 27 February 2025. A training collaboration, not a standard or exposure draft.

[26] Organisation for Economic Co-operation and Development. *AI Incidents and Hazards Monitor.* OECD.AI. https://oecd.ai/en/incidents

[27] Responsible AI Collaborative. *AI Incident Database.* https://incidentdatabase.ai

One verification could not be completed. No trademark register was reachable from the environment in which this registry's checks were run, and no ownership claim regarding any third-party mark is therefore made anywhere in this document. Section 4.6.2 states the distinction Five-Gate draws without asserting who owns the mark it distinguishes itself from. That is deliberate, and it is the correct form for a document that cannot verify the assertion.

Appendix A. Former Names and Retired Marks

Recorded so that work published under a former name remains citable. **A former name is not an error in the work that used it.**

A.1 The twelve marks as asserted

Asserted on the copyright page of the Enterprise Playbook, First Edition, 2026 [13]: MESA Framework, MESA MRM Framework, AI Data Governance Stack, AI Vendor Risk Framework (AVRF), AI Incident Response Protocol (AIRP), Governance Office Blueprint, RACI-AI Matrix, Five-Gate Deployment Model, Governance Cadence Framework, Governance Maturity Model, Sharia AI Compliance Framework (SACF), Cross-Border AI Architecture Patterns.

A.2 Renames

Former name	Current canonical name	Recorded at
AI Data Governance Stack	AI Data Governance Framework™ (REG-03)	6.4
AI Governance Operating Model	CADRE™ (REG-07)	6.9

The second rename is not like the first. *AI Data Governance Stack* is withdrawn from use as a name and survives only as the correct citation for work published under it. *AI Governance Operating Model* is **retired as the name and retained as the descriptor**: it continues in current use, in the canonical first mention *CADRE™*,

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the AI Governance Operating Model, and a document using it that way is current rather than historical. It is listed here because it was a canonical name and a reader tracing one will look here first.

A.3 Consolidated names

Former name	Became	Recorded at
Governance Office Blueprint	The Structure section of REG-07	6.2
RACI-AI Matrix	The Decision rights section of REG-07	6.2
Governance Cadence Framework	The Cadence section of REG-07	6.2

A.4 Retired instruments

Name	Status	Recorded at
Governance Maturity Model	Superseded by the MESA maturity vector	6.3

A.5 Retired acronym expansion

Former expansion	Current expansion	Recorded at
MESA: Middle East Strategic Alignment	MESA: Maturity, Evidence, Substrate, Alignment	4.1.2

The mark itself is unchanged. Only the expansion is retired, and the framework's origin is a matter of record that is not disclaimed.

A.6 Names that are deliberately unmarked

Name	Family	Why unmarked
The Boundary Invariant	Engineering	Adoption is the strategy; a mark would suppress the citation. See 2.3 and [2]
The BOE Declaration	Engineering	A document convention meant to be copied. See 2.3 and [2]
PEVG, PARA	Engineering	Pattern names spread; marked pattern names do not. See [2]

The governance family has no deliberately unmarked name at version 2.0. *AI Governance Operating Model* stood in this table at version 1.0 and was removed when REG-07 was renamed and marked, recorded at 6.9. The reasoning that put it here is not withdrawn and is rewritten at 4.7.4, because the question it recorded as open is now closed and the record of why it was open is part of the answer.

Appendix B. Publication Provenance

B.0 What version 2.0 is. The second public version, superseding version 1.0 under the same concept DOI, which continues to resolve to the latest version. **It is a major version for one reason**, the rename of REG-07 recorded at 6.9, and 11.1 classifies a canonical name change as major because a claim written against the former name may no longer be conformant against the new one. Everything else in this version is minor-class change carried in the same release: five statuses closed, five entries reconciled with their specifications, two routed questions disposed of and one correction to version 1.0. **Every change is classified in `CHANGELOG.md` as 11.2 requires.** The nine entries, the thirty-one edges, the relationship vocabulary and the conformance requirements of Section 8 are unchanged, so a conformance claim made against version 1.0 remains valid at version 2.0 unless it named

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REG-07 by its former name, which A.2 and 6.9 address. The paragraphs below are retained from version 1.0 and describe the registry's first publication.

B.1 What this document is. The first public version of the Defensible AI Framework Registry. It has not been peer reviewed. It is deposited on Zenodo, which confers a DOI, a timestamp and citability. **Zenodo does not confer standard status, and nothing in this document uses the word standard for itself.**

B.2 Relationship to prior publication. The nine frameworks were published individually before this registry, principally in [13]. This document is the first statement of them as one architecture, the first statement of the relationships between them, and the first dated instrument of the consolidation recorded in Section 6.

B.3 No preprint. This registry carries no preprint identifier, and no relation to any preprint is asserted in its metadata, because none exists.

B.4 Publication order. This registry is published after version 1.1 of [1] and version 1.1 of [2]. The order is a requirement rather than a convenience: the MESA entry at 4.1.2 carries an acronym expansion adopted in [1] version 1.1, and Section 2 rests on the Boundary Invariant as named in [2] version 1.1. Both references above are concept DOIs, which always resolve to the latest version, so this document does not go stale when either is versioned again.

B.5 Verification method. Every external reference in Section 12 was checked against a live primary source with a deliberate failing control, on 29 August 2026. Where a source could not be reached in a way that discriminated a real identifier from an invented one, no claim resting on it appears in this document. The one such case is recorded in the note at the end of Section 12.

B.6 Funding and conflicts. The work is unfunded. The author operates a consulting practice applying these frameworks, which is a conflict of interest stated here and addressed at 10.7.

Document History

Version	Date	Summary
1.0	30 August 2026	First public version. Nine entries, thirty-one relationship edges, the consolidation notice, and the machine-readable layer
2.0	15 September 2026	Major version. REG-07 renamed to CADRE and marked; all five remaining `source-treatment` entries move to `deposited` and are reconciled with their specifications; the gate five supply edge and the authority grant to REG-09 disposed of; one correction to 11.4

About the Author

Nabeel Khan is an enterprise AI architect and governance advisor, and Founder & Solutions Architect at iSystematic Inc. He has spent twenty-five years architecting enterprise data and machine-learning systems in government, healthcare, financial services and telecommunications across three continents, grounded in TOGAF, DMBOK, ISO 27001 and SOC 2.

He works on governed production AI; systems whose authority is bounded by the architecture, whose decisions carry the evidence needed to defend them, and whose behaviour the institution can explain afterwards. The frameworks specified here, and the regulatory knowledge behind them, are published as machine-accessible AI expertise and are callable over MCP without a key at mcp.nabeelkhan.com/api/mcp, so a reader may query this specification's contents directly and receive answers carrying their source.

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```
Khan, N. A. (2026). The Defensible AI Framework Registry: Canonical
  Names, Definitions and Relationships for the Governed Production AI
  Discipline.
(Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22170112
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Trademarks. No trademark is claimed on the name of this registry. Eight of the nine frameworks it catalogues are marked, and those marks are listed in the trademark notice above. The CC BY licence covers this text and grants no rights in any mark.